EEE 311, ETE 311 lab: Electronics II

Experiment No: 02

Name of the Experiment:

Study of basic comparator circuits using op-amp

Objectives:

1. Study the basics of op-amp.
2. Study the use of op-amp as a comparator.
3. Study the op-amp voltage transfer characteristics.

Theory:

The operational amplifier (abbreviated as OP-AMP) is a direct-coupled high-gain amplifier to which feedback is added to control its overall response characteristic. It has very high (ideally ∞) input impedance and very low (ideally 0) output impedance. It offers all advantages of monolithic integrated circuits, such as small size, high reliability, reduced cost, and low offset voltage and current. For these reasons, it has gained wide acceptance as a versatile, predictable and economic system building block. The pin diagram of dual in-line package IC-741 is shown in Figure 1 along with the op-amp circuit symbol and connections to the pins.

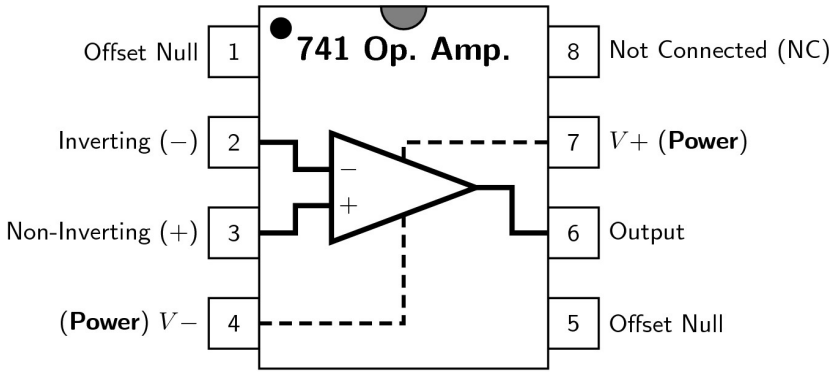


Figure 1: Op-amp (IC-741) pin diagram for dual in-line package (DIP or DIL)

Comparator Unit Operation:

The Op-amp comparator compares one analogue voltage level with another analogue voltage level, or some preset reference voltage, V_{ref} and produces an output signal based on this voltage comparison. In other words, the op-amp voltage comparator compares the magnitudes of two voltage inputs and determines which is the largest of the two. The op-amp comparator studied in this experiment does not contain any feedback path. Hence, this is called *open-loop comparator*. In the open-loop configuration, op-amp provides its very large *open-loop gain* (A_{OL}), which is ideally ∞ and practically $10^{5\text{-}6}$.

A basic comparator circuit can be represented as in Figure 2a. The output is a digital signal that stays at a high voltage level (V_{high}) when the non-inverting input (V_+) is greater than the voltage at the inverting input (V_-) and switches to a lower voltage level (V_{low}) when the non-inverting input voltage goes below the inverting input voltage.

$$V_+ > V_-: V_o = V_{high}$$
$$V_- > V_+: V_o = V_{low}$$

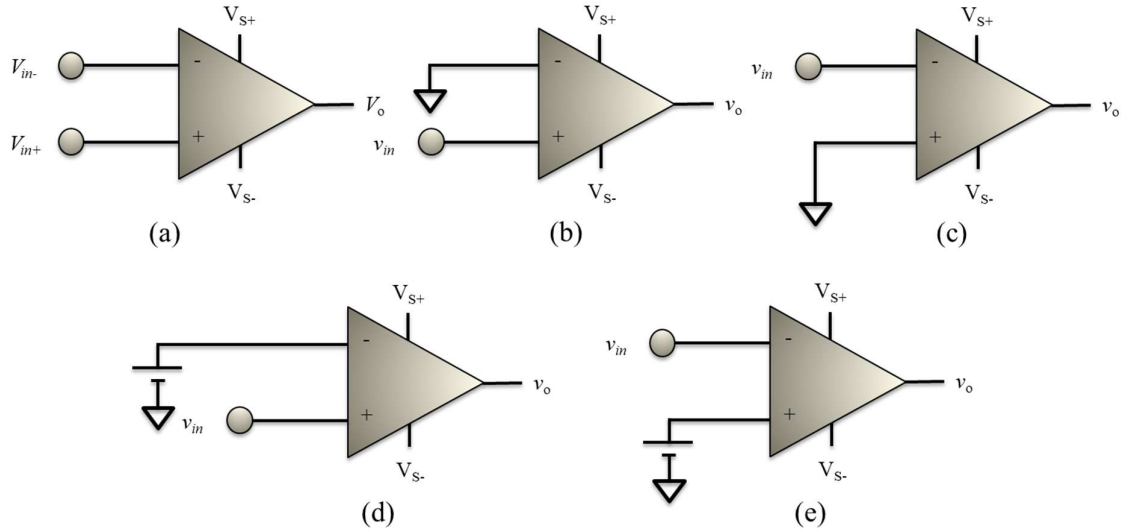


Figure 2. Op-amp open-loop comparator circuits, comparing (a) 2 DC input signals, (b, c) an AC signal with ground reference, (d, e) an AC signal with a positive DC reference. V_{s+} and V_{s-} refer to the positive and negative power supply voltages.

Figure 2b shows a typical open loop comparator. The inverting input in this example is connected to a ground reference voltage, while the non-inverting input is connected to the input signal voltage. If v_{in} is less than the reference voltage level of 0 V, the output remains at V_{low} (near $V_{s-} = -10\,V$). When the v_{in} rises just above 0 V, the output quickly switches to V_{high} level (near $V_{s+} = +10\,V$). This type of circuit configuration is known as the *non-inverting open-loop comparator*. In an *inverting open-loop comparator* (Figure 2c), the input signal is connected to the inverting input terminal, whereas the reference voltage is connected to the non-inverting input terminal. When we compare a signal with a 0 V reference, we also call these circuits *zero crossing detector*s (Figure 2b, c). When we compare a signal with a positive DC voltage reference, we call these circuits *positive level indicator*s. Figures 2d and 2e show non-inverting and inverting positive level indicators, respectively. Similarly, *negative level indicator*s can be constructed by connecting negative DC voltage sources at the reference input terminals.

The internal circuit of an op-amp contains differential and push-pull amplifier stages resulting in a high A_{OL}. Hence, input signals of any shape (sinusoidal, triangular, or others) in the V, mV, or even hundreds of μV range leads to saturation in the output signal. Figure 3a and 3b show saturated rectangular output signals for the non-inverting and inverting zero-crossing detectors for the sinusoidal input signal. With reference input (at pin 2) set to 0 V, a sinusoidal signal applied to the non-inverting input (pin 3) will cause the output of the non-inverting comparator to switch between its two output states without any phase difference, as shown in Figure 3a. For the inverting configuration of the open-loop comparator, on the other hand, a similar output is obtained with a 180° phase difference, as show in Figure 3b.

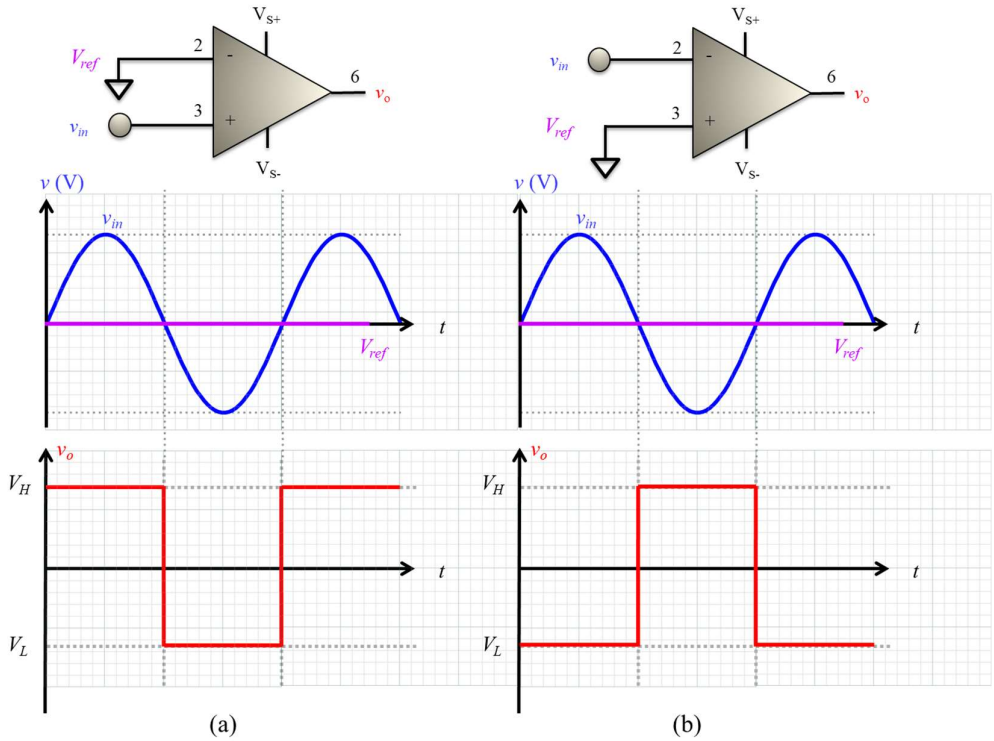


Figure 3: The input signal (v_{in}), reference voltage (V_{ref}), and resulting output signal (v_{out}) of open-loop zero crossing detector comparator circuits with (a) non-inverting configuration, (b) inverting configuration.

For the non-inverting configuration, when input v_{in} goes even a fraction of a millivolt above the 0 V reference level, the voltage difference will be amplified by the very A_{OL} so that the output rises to its positive output saturation level and remains there while the input stays above $V_{ref} = 0$ V. When the input drops just below the 0 V reference level, the output is driven to its lower saturation level and stays there while the input remains below $V_{ref} = 0$ V. Figure 4a and 4b show the transfer characteristic, v_o vs. v_{in}, for the non-inverting and inverting zero-crossing detectors, respectively. In the narrow linear regions, the v_o follows v_{in} with a constant slope of $+A_{OL}$ (non-inverting) and $-A_{OL}$ (inverting). In the saturation regions, the v_o is at V_{s+} when v_{in} is positive and at V_{s-} when v_{in} is negative for the non-inverting configuration. This is opposite for the inverting configuration.

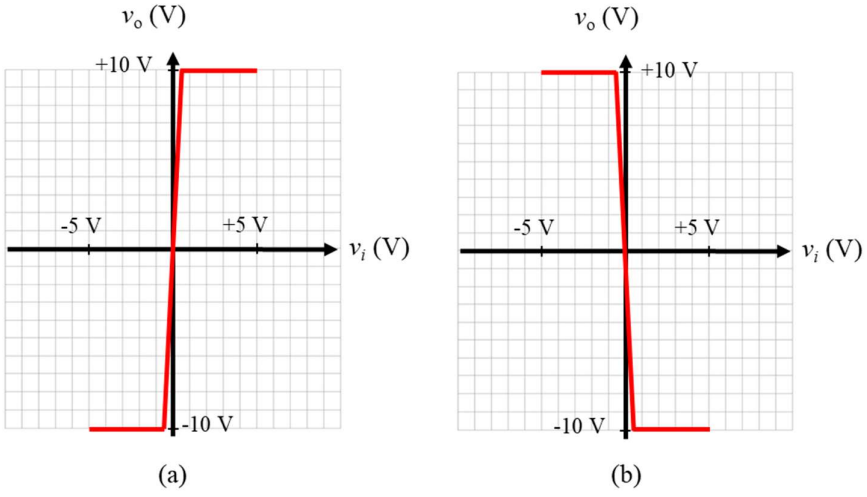
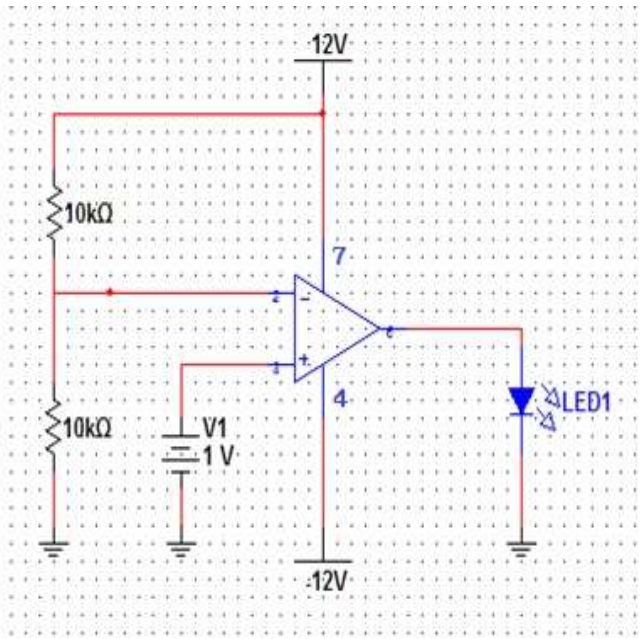


Figure 4. Transfer characteristic of open-loop comparator of (a) non-inverting and (b) inverting configurations.

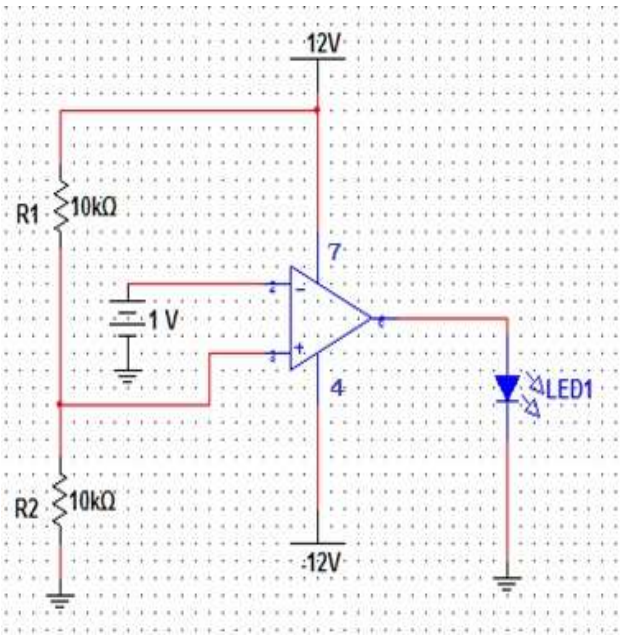
Equipment List:

Serial No.	Component Details	Specification	Quantity
1.	OPAMP	741	1 piece
2.	Resistor	10 kΩ, 5 kΩ	2 pieces, 1 piece
3.	LED	Blue	1 piece
4.	Oscilloscope		1 unit
5.	DC Power Supply		3 units
6.	Function Generator		1 unit
7.	Oscilloscope	2 channels	1 unit
8.	Digital Multimeter		1 unit
9.	Trainer Board		1 unit

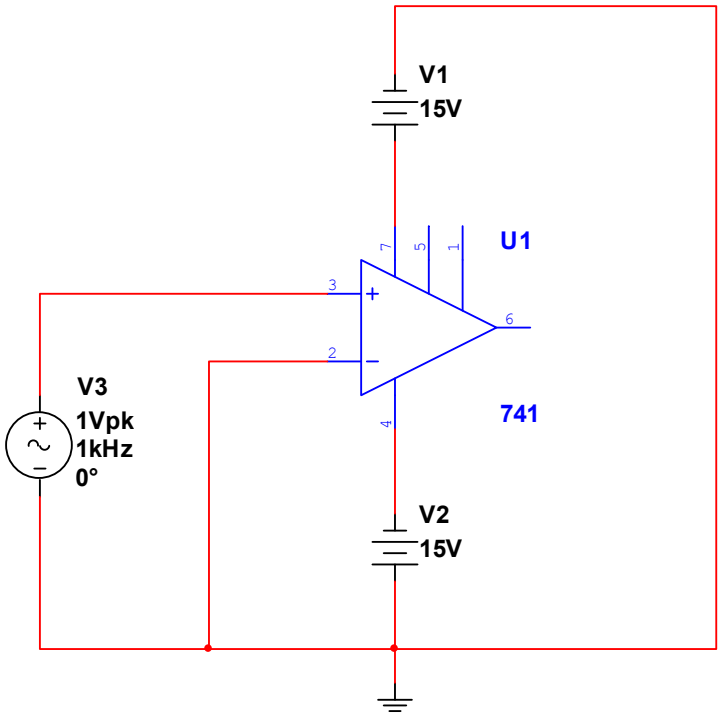
Circuit Diagrams:



Circuit 1 – Open-loop comparator 1: Constant DC input at +ve input terminal and a constant DC reference voltage via a voltage divider network at the -ve input terminal.



Circuit 2 – Open-loop comparator 2: Constant DC input at -ve input terminal and a constant DC reference voltage via a voltage divider network at the +ve input terminal.



Circuit 3 – Open-loop non-inverting comparator as zero crossing detector: AC input at +ve input terminal and a constant DC 0 V reference voltage (ground) at the -ve input terminal.

Procedure:

Circuit 1

- Connect the circuit shown in circuit 1.
- Set the biasing voltage V_{CC} = 12 V at pin 7 and V_{EE} = -12 V at pin 4.
- Set input DC voltage to 0 V. Check the response of LED (on/off).
- Now increase the input voltage until the state of LED change. Take readings.

V_{ref}	V_1	State of LED
	0 V	
	1 V	
	2 V	
	3 V	
	4 V	
	5 V	
	6 V	
	7 V	
	8 V	

Circuit 2

- Connect the circuit shown in circuit 2.
- Set the biasing voltage V_{CC} = 12 V at pin 7 and V_{EE} = -12 V at pin 4.
- Set input DC voltage to 0 V. Check the response of LED (on/off).
- Now increase the input voltage until the state of LED change. Take readings.

V_{ref}	V_1	State of LED
	0 V	
	1 V	
	2 V	
	3 V	
	4 V	
	5 V	
	6 V	
	7 V	
	8 V	

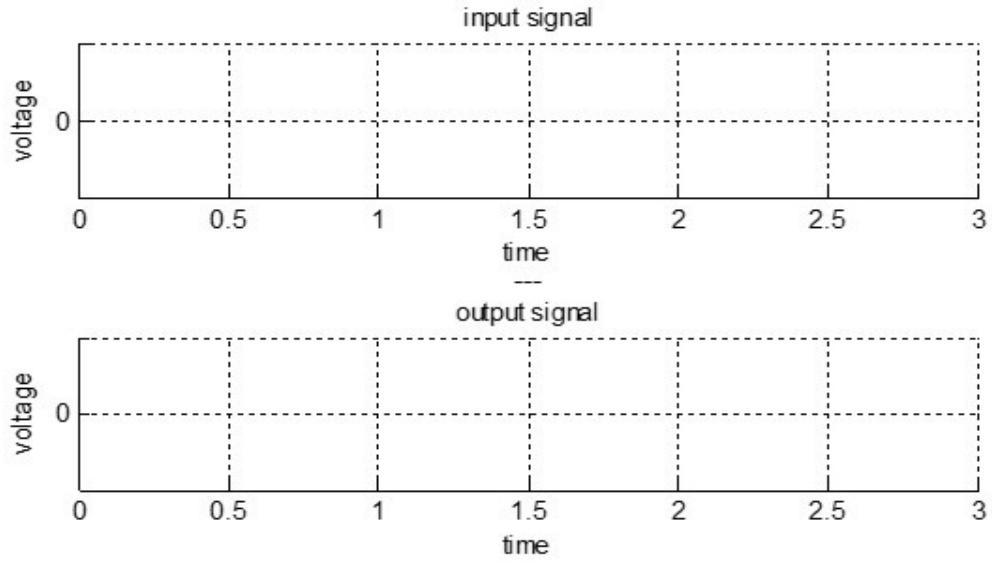
- Change the value of R_2 to 5 kΩ. Repeat previous steps.

V_{ref}	V_1	State of LED
	0 V	
	1 V	
	2 V	
	3 V	
	4 V	
	5 V	
	6 V	
	7 V	
	8 V	

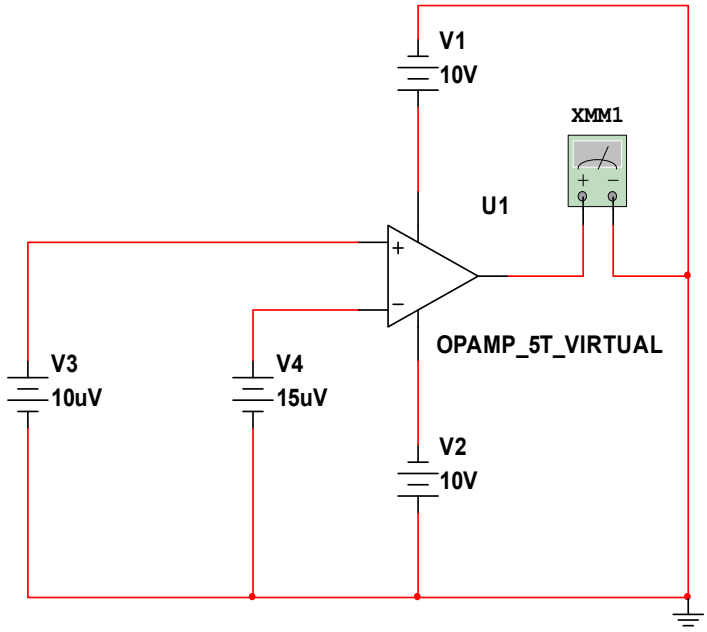
Circuit 3

- Make the connection as shown in circuit 3.
- Apply an input **sine wave** with a peak-to-peak voltage = 2 V and frequency = 1 kHz.
- Observe the input and output wave shapes through an oscilloscope.
- Take reading of its amplitude. $V_{in(p\text{-}p)}$ = _____, $V_{out(p\text{-}p)}$ = _____
- Draw the input and output wave shapes with proper labeling.

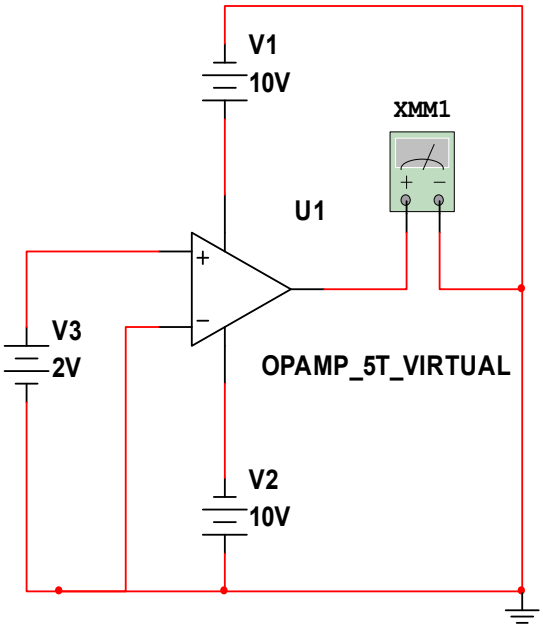
Graphs:



MULTISIM based lab work:



Circuit – 4



Circuit – 5

Procedure:

Circuit 4

- Construct the circuit in MULTISIM.
- Set input offset voltage (V_{OS}) to 0 V and input offset current (I_{OS}) to 0 A.
- Set to open loop gain (A) to 1 M V/V
- Set V3 and V4 and complete the following table.

V3	V4	V_d	V_O	Liner / Saturation
-15 µV	-10 µV			
+15 µV	-10 µV			
1.000000 V	1.000001 V			
+6 mV	+2 mV			
0	+1mV			

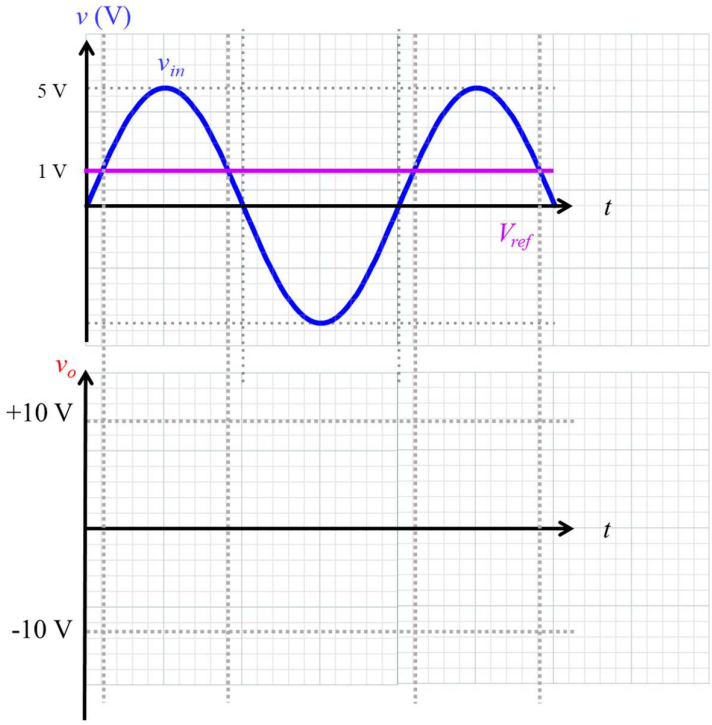
Circuit 5

- Construct the circuit in MULTISIM.
- Set input offset voltage (V_{OS}) to 0 V and input offset current (I_{OS}) to 0 A.
- Set to open loop gain (A) to 100,000 V/V
- Take parameter sweep (or DC sweep) for V3

- Start from -1 mV and stop at 1 mV. For better accuracy, take at least 1000 data points.
- Using the output plot, determine the open loop gain.

Questions:

1. What are the applications of comparators? How can we build analog-to-digital converters using the concept of comparators?
2. How can we use voltage dividers to change the reference voltage of a comparator?
3. Draw the output v_o waveform for the following input signal v_{in} and reference V_{ref} for (i) non-inverting configuration, (ii) inverting configuration. Use V_{s+} of +10 V and V_{s-} of -10 V. $A_{OL}=10^6$.



4. What is a window comparator? How many op-amps do you need for a window comparator? How many reference voltages are present in a window comparator?

Resources:

Check the following web link for further information on comparators. You can also find information on the window comparator here:
https://www.electronics-tutorials.ws/opamp/op-amp-comparator.html

EEE 311, ETE 311, CSE 351 lab: Electronics II

Experiment No: 03

Name of the Experiment:

Inverting Amplifier, Non-Inverting Amplifier, and Voltage Follower Circuits of Op-amps

Objectives:

a. To study the basic operation of an inverting amplifier using op-amp.
b. To study the basic operation of a non-inverting amplifier using op-amp.
c. To study a voltage follower using op-amp.

Theory:

- **Inverting Amplifier:**

An inverting amplifier is a special case of the closed-loop amplifier in which the op-amp's non-inverting input terminal is grounded and inverting input terminal is connected to an input voltage source via an input resistance. A time-varying ac input signal is 180° phase shifted at the output, whereas a constant dc signal has a reversal of polarity at the output.

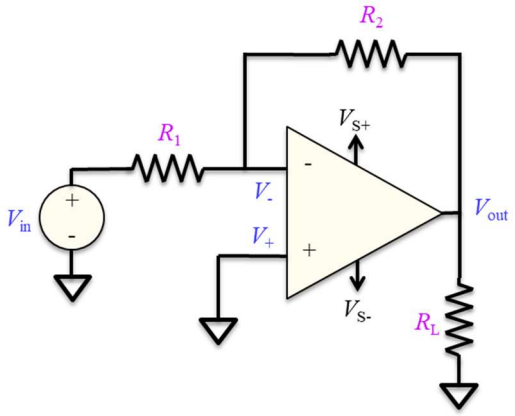


Figure 1: A typical inverting amplifier circuit.

Figure 1 shows a typical inverting amplifier circuit. In this circuit, there is an input voltage source V_{in} near the inverting input terminal of the op-amp. By applying KCL at the inverting input terminal, we get:

$$\frac{V_{in} - V_-}{R_1} + \frac{V_{out} - V_-}{R_2} = 0$$

In a closed-loop circuit constructed with op-amp, $V_- = V_+$ due to virtual short circuit (where V_- and V_+ are the node voltages at the inverting and non-inverting input terminals, respectively). In this circuit, $V_- = V_+ = 0$ V due to virtual ground. By replacing V_- with 0 in our KCL equation, we get:

$$\frac{V_{in}}{R_1} + \frac{V_{out}}{R_2} = 0 \quad \Rightarrow \quad \frac{V_{out}}{R_2} = -\frac{V_{in}}{R_1} \quad \Rightarrow \quad V_{out} = -\left(\frac{R_2}{R_1}\right) V_{in}$$

$$A_{CL} = \frac{V_{out}}{V_{in}} = -\frac{R_2}{R_1}$$

The input resistance of the inverting amplifier is limited by the resistance R_1.

- **Non-inverting Amplifier:**

Figure 2 shows a typical non-inverting amplifier circuit. A non-inverting amplifier is another special case of closed-loop amplifier in which the op-amp's non-inverting input is connected to the input voltage source. A time-varying ac input signal remains in phase at the output, whereas a constant dc signal has the same polarity at the output.

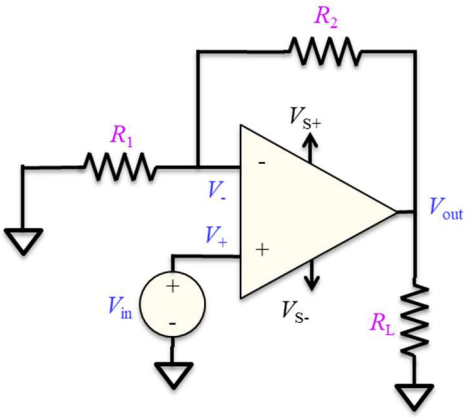


Figure 2: A typical non-inverting amplifier circuit.

By applying KCL at the inverting input terminal, we get:

$$\frac{V_- - 0}{R_1} = \frac{V_{out} - V_-}{R_2}$$

In a closed-loop circuit constructed with op-amp, $V_- = V_+ = V_{in}$ due to virtual short circuit. Hence, we can simplify the above equation as,

$$\frac{V_{in}}{R_1} = \frac{V_{out}}{R_2} - \frac{V_{in}}{R_2} \quad \Rightarrow \quad \frac{V_{out}}{R_2} = V_{in}\left(\frac{1}{R_1} + \frac{1}{R_2}\right) \quad \Rightarrow \quad V_{out} = \left(1 + \frac{R_2}{R_1}\right) V_{in}$$

$$A_{CL} = \frac{V_{out}}{V_{in}} = 1 + \frac{R_2}{R_1}$$

The input resistance of the non-inverting amplifier is ideally infinite and practically very large.

- **Voltage Follower/ Buffer:**

A voltage follower produces almost the same output as input. The input resistance of a voltage follower circuit is very high (several mega ohms practically). Therefore, it draws negligible current from a signal source. Thus, it works as a voltage buffer that provides a means of isolating an input signal from a load.

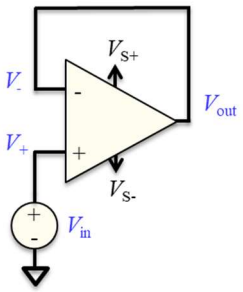


Figure 3: A voltage follower/ buffer circuit.

We know, for a non-inverting amplifier, the gain is:

$$A_{CL} = \frac{V_{out}}{V_{in}} = 1 + \frac{R_2}{R_1}$$

For this non-inverting voltage follower, $R_2 = 0$, $R_1 = \infty$.

$$A_{CL} = \frac{V_{out}}{V_{in}} = 1 + \frac{0}{\infty} \quad \Rightarrow \quad A_{CL} = \frac{V_{out}}{V_{in}} = 1 \quad \Rightarrow \quad V_{out} = V_{in}$$

Like the unity-gain non-inverting voltage follower, inverting unity-gain voltage follower is also possible. The magnitude of the output is theoretically equal to the input, but the output will be 180° phase shifted. The input resistance of the inverting unity-gain voltage follower is limited.

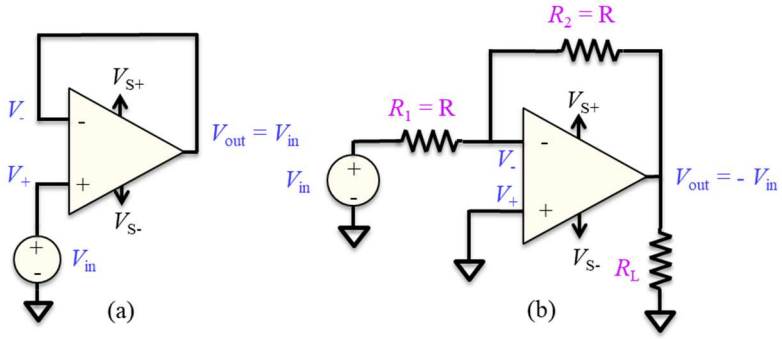


Figure 4: (a) Unity-gain non-inverting voltage follower circuit, (b) Unity-gain inverting voltage follower circuit

Equipment List:

	Component Details	Specification	Quantity
1.	Op-amp	LM741	1 piece
2.	Resistor	10 kΩ, 100 kΩ	2 pieces, 1 piece
3.	Oscilloscope		1 unit
4.	AC function generator		1 unit
5.	DC Power Supply		3 units
6.	Digital Multimeter		1 unit
7.	Trainer Board		1 unit

Circuit Diagrams:

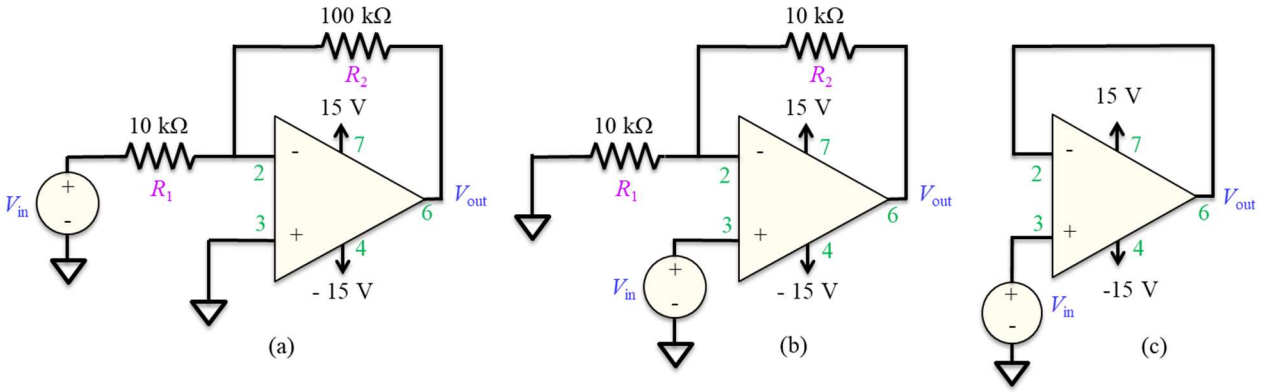
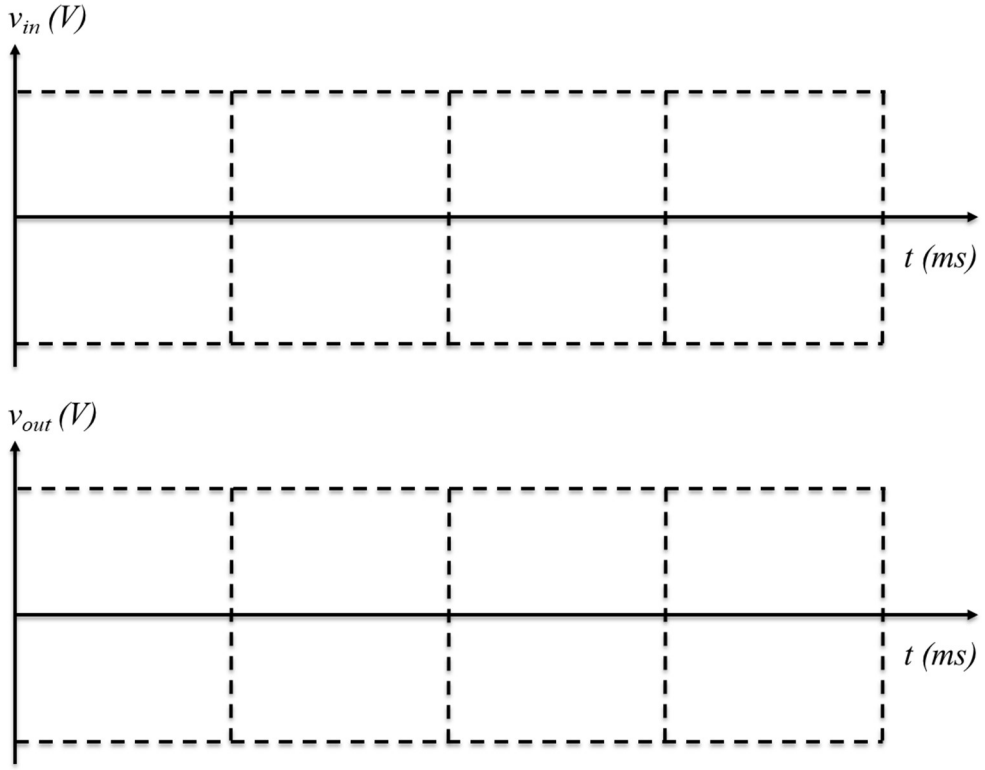


Figure 5. Circuit implementations: (a) Circuit 1 - Inverting Amplifier, (b) Circuit 2 - Non-Inverting Amplifier, (c) Circuit 3 - Voltage Follower

Procedure:

Study of inverting amplifier:

1. Construct circuit 1. Set $V_{S+} = 15$ V and $V_{S-} = -15$ V.
2. Apply a **sine wave** with a peak-to-peak voltage = 0.5 V and frequency = 1 kHz as the input.
3. Find out V_{out} (by Oscilloscope) = _____



4. Calculate the gain A_{CL} and the phase difference

A_{CL} = _____. $\Delta\theta$ = _____.

5. Replace R_2 with a 10 kΩ resistor and repeat all the processes.
6. Find out V_{out} (by Oscilloscope) = _____
7. Calculate the gain A_{CL} and the phase difference

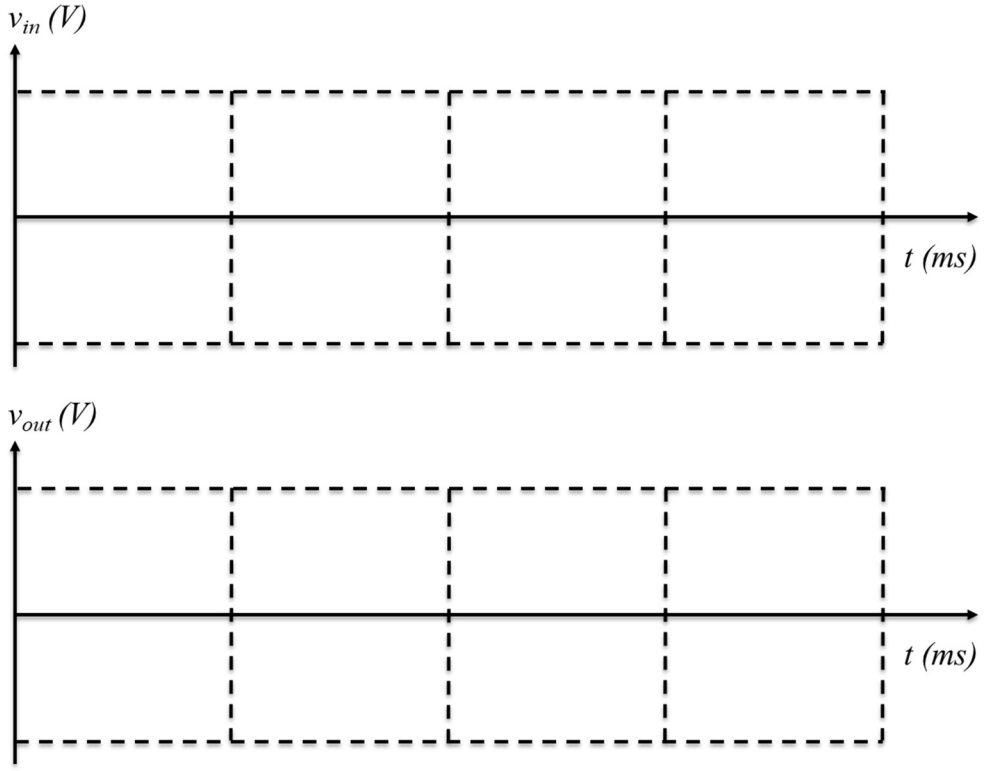
   A_{CL} =_____. $\Delta\theta$ = _____.

8. Now replace the function generator with a dc power supply and set V_{in} = 2 V (DC).
9. Find out V_{out} (by Oscilloscope or DMM) = _____
10. Calculate the gain A_{CL} and the phase difference

    A_{CL} =_____. $\Delta\theta$ = _____.

Study of non-inverting amplifier:

1. Construct circuit 2. Set V_{S+} = 15 V and V_{S-} = –15 V.
2. Apply a **sine wave** with a peak-to-peak voltage = 0.5 V and frequency = 1 kHz as the input.
3. Find out V_{out} (by Oscilloscope) = _____



4. Calculate the gain A_{CL} and the phase difference

   A_{CL} =_____. $\Delta\theta$ = _____.

5. Now replace the function generator with a dc power supply and set V_{in} = 2 V (DC).
6. Find out V_{out} (by Oscilloscope or DMM) = _____
7. Calculate the gain A_{CL} and the phase difference

   A_{CL} =_____. $\Delta\theta$ = _____.

Study of the voltage follower:

1. Construct circuit 3. Set V_{S+} = 15 V and V_{S-} = –15 V.

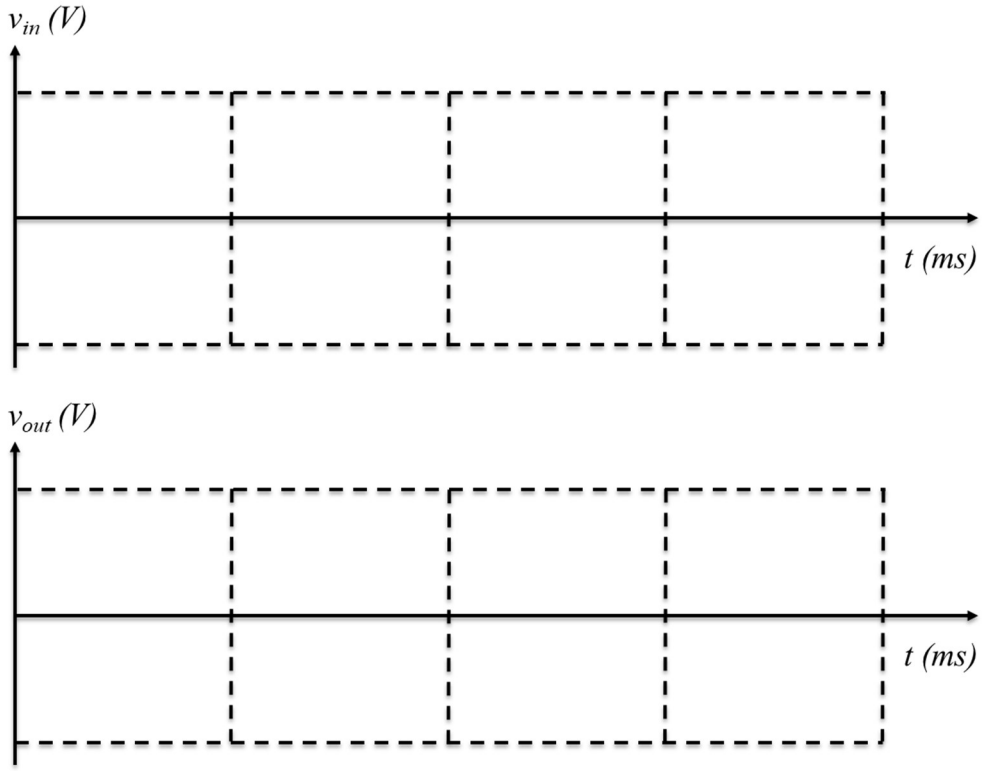
2. Measure V_{out} (by DMM) for V_{in} = +5 V (DC)

   V_{out} = _____.

3. Measure V_{out} (by DMM) for V_{in} = –5 V (DC)

   V_{out} = _____.

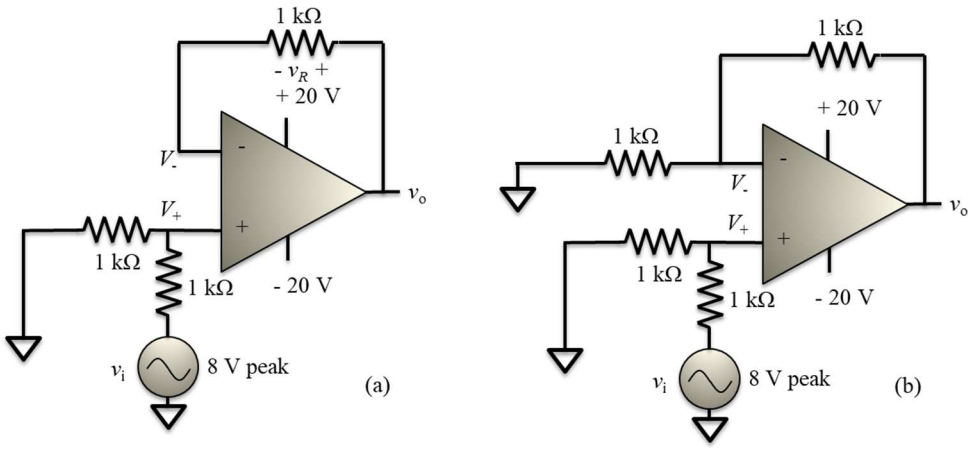
4. Now replace the dc power supply with a function generator.
5. Apply a **sine wave** with a peak-to-peak voltage = 2 V and frequency = 1 kHz as the input.
6. Measure V_{out} (by Oscilloscope) = _____.



7. Find out the phase difference between V_{in} and V_{out}. $\Delta\theta$ = _____.

Questions:

1. How does the unity-gain voltage follower isolate a load from a bad source including a large source resistance of 1 MΩ?
2. What are the advantages of closed-loop circuits over open-loop circuits?
3. Derive the output voltage v_o for the following circuits:



4. What is the bandwidth of an ideal op-amp and practical op-amp (LM 741c)?
5. What are the output resistances of the inverting amplifier, non-inverting amplifier, and voltage follower?
6. What are the input resistances of the inverting amplifier, non-inverting amplifier, and voltage follower?

EEE 311, ETE 311, CSE 351 lab: Electronics II

Experiment No: 04

Name of the Experiment:

Adder and Subtractor Circuits using Op-amps

Objectives:

a. To study the operation of an inverting adder circuit using op-amp.
b. To study the operation of a differential amplifier / subtractor circuit using op-amp.

Theory:

- **Inverting Adder:**

Inverting adder adds up the signals at its inverting input terminal and produces the inverse of this summation at the output when the value of the feedback resistance and the resistances of the resistors connected in series with the input signals are equal. When these resistors are unequal, we construct either a weighted adder or a summing amplifier.

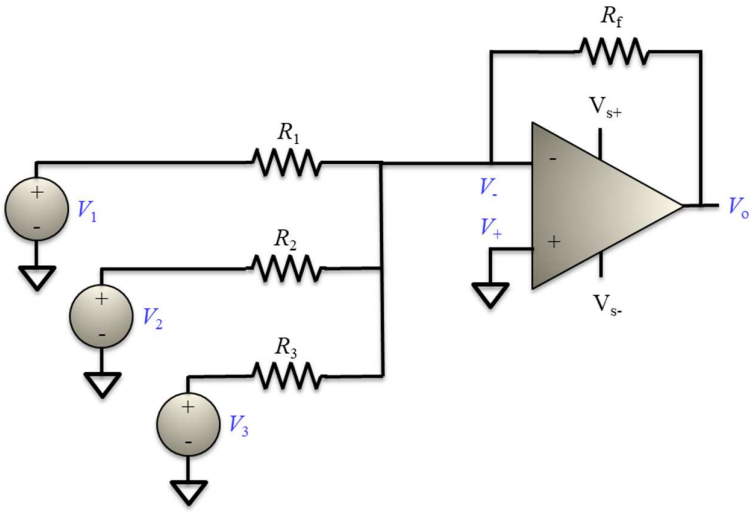


Figure 1: A typical inverting weighted adder circuit.

In this circuit, all the input voltages are connected at the negative terminal of the op-amp. If the node voltage of the negative terminal is $V_$, applying KCL at that node we get:

$$\frac{V_1 - V_-}{R_1} + \frac{V_2 - V_-}{R_2} + \frac{V_3 - V_-}{R_3} + \frac{V_o - V_-}{R_f} = 0$$

Due to the virtual short circuit of closed-loop configuration of ideal op-amp, $V_- = V_+$ (where V_+ is the node voltage at the non-inverting input terminal). Due to the virtual ground of the inverting closed-loop amplifier, $V_- = V_+ = 0$. Hence, the above equation can be simplified as:

$$\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} + \frac{V_o}{R_f} = 0$$

The output voltage expression of the weighted adder is deduced from the above equation as,

$$V_o = \left(-\frac{R_f}{R_1}V_1 - \frac{R_f}{R_2}V_2 - \frac{R_f}{R_3}V_3\right)$$

$$\Rightarrow V_o = -R_f\left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}\right)$$

We construct a summing amplifier when $R_1 = R_2 = R_3 = R_i$. The output voltage expression of a summing amplifier is simplified from the above equation as,

$$V_o = -\frac{R_f}{R_i}(V_1 + V_2 + V_3)$$

We construct an inverting adder when $R_f = R_1 = R_2 = R_3$. The output voltage expression of an inverting adder is simplified from the above equation as,

$$V_o = -(V_1 + V_2 + V_3)$$

- **Differential Amplifier and Subtractors:**

Differential amplifiers amplify the difference between two input voltages. This is a simplified version of a weighted subtractor circuit. An even further simplified version is a subtractor circuit.

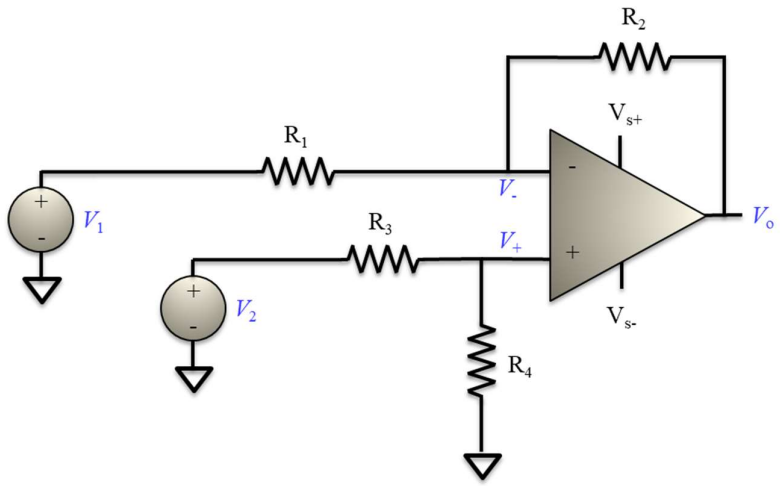


Figure 2: A typical weighted subtractor circuit.

Here, one input signal V_1 is connected to the inverting input terminal (pin 2) while the other input signal V_2 is connected to the non-inverting input terminal (pin 3). Due to the virtual short circuit of closed-loop configuration of ideal op-amp, $V_- = V_+$. Applying KCL at the negative input terminal,

$$\frac{V_1 - V_-}{R_1} = \frac{V_- - V_o}{R_2}$$

$$\Rightarrow\ V_o = V_- - \frac{R_2}{R_1}(V_1 - V_-)$$

Using voltage divider rule, we get

$$V_+ = \frac{R_4}{R_3 + R_4} V_2$$

From the above two equations, we get

$$V_o = \left(\frac{R_4}{R_3 + R_4}\right) V_2 - \frac{R_2}{R_1}(V_1 - \frac{R_4}{R_3 + R_4} V_2)$$

$$V_o = -\left(\frac{R_2}{R_1}\right) V_1 \ + \ \left(\frac{R_4}{R_3 + R_4}\right)\left(1 + \frac{R_2}{R_1}\right) V_2$$

The above equation shows the output voltage expression of a weighted subtractor circuit. We can simplify this by using $R_1 = R_3$ and $R_2 = R_4$ to construct a difference or differential amplifier circuit,

$$V_o = \ -\left(\frac{R_2}{R_1}\right) V_1 + \left(\frac{R_2}{R_1}\right) V_2$$

$$V_o = \left(\frac{R_2}{R_1}\right)\ (V_2 - V_1)$$

The circuit will operate as a differential amplifier and $\left(\frac{R_2}{R_1}\right)$ will be the differential gain A_D. We can further simplify this to construct a subtractor circuit by using $R_1 = R_2 = R_3 = R_4$,

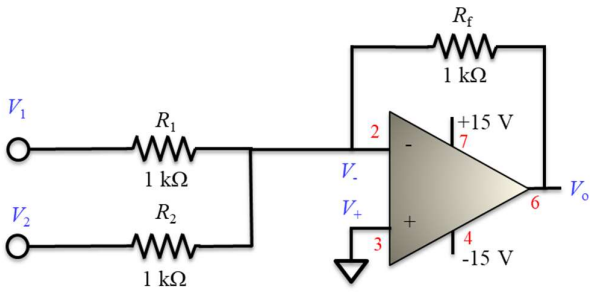
$$V_o = \ -V_1 + V_2$$

$$V_o = \ V_2 - V_1$$

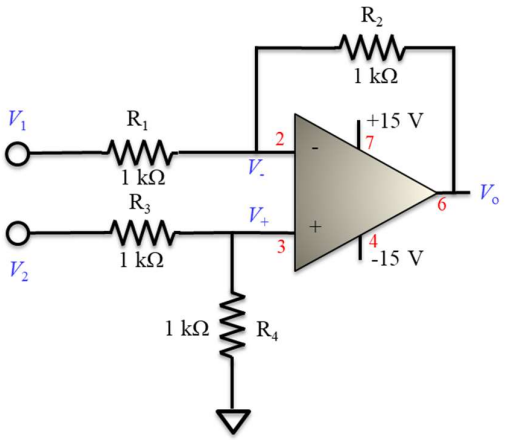
Equipment List:

	Component Details	**Specification**	**Quantity**
1	Op-Amp	LM741	1 piece
2	Resistor	1 kΩ, 3.3 kΩ	4 pieces, 2 pieces
3	DC Power Supply		3 units
4	AC function generator		1 unit
5	Oscilloscope		1 unit
6	Digital Multimeter		1 unit
7	Breadboard		1 unit

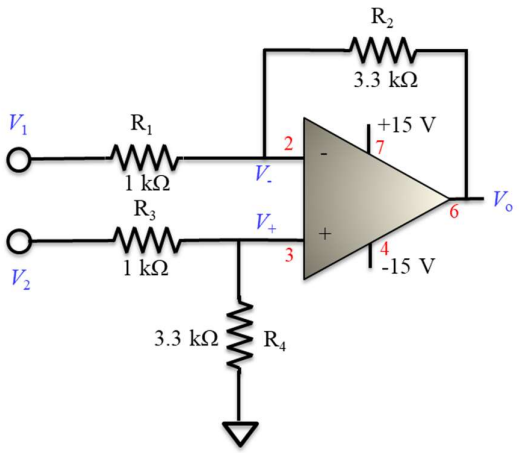
Circuit Diagrams:



Circuit 1 - Inverting Adder



Circuit 2 - Subtractor



Circuit 3 – Difference Amplifier

Procedure:

Study of inverting Adder:

1. Construct the Circuit 1. Set $V_{S+} = 15$ V and $V_{S-} = -15$ V.
2. Obtain the values of V_o and complete the following table.

V_1	V_2	V_o (theoretical)	V_o (practical)
5 V	1 V	-6 V	
5 V	2 V	-7 V	
5 V	3 V	-8 V	
5 V	4 V	-9 V	
5 V	5 V	-10 V	
5 V	-3 V	-2 V	

3. Now connect a function generator as v_2. Keep the frequency at 1 kHz. Obtain values of v_o and complete the following table.

V_1 (DC)	v_2 (AC)	$v_{o\ (p\text{-}p)}$	$V_{o\ (max)}$	$V_{o\ (min)}$
5 V	1 $V_{p\text{-}p}$			
5 V	2 $V_{p\text{-}p}$			
5 V	3 $V_{p\text{-}p}$			

4. Calculate the percentage of error.

Study of Subtractor:

1. Construct the Circuit 2. Set V_{S+} = 15 V and V_{S-} = –15 V.
2. Obtain the values of V_o and complete the following table.

V_1	V_2	V_o (theoretical)	V_o (practical)
1 V	5 V	4 V	
2 V	5 V	3 V	
3 V	5 V	2 V	
4 V	5 V	1 V	
5 V	5 V	0 V	
-3 V	5 V	8 V	

3. Calculate the percentage of error.

Study of Difference Amplifier:

1. Construct the Circuit 3. Set V_{S+} = 15 V and V_{S-} = –15 V.
2. Obtain the values of V_o and complete the following table.

V_1	V_2	V_o (practical)
1 V	5 V	
2 V	5 V	
3 V	5 V	
4 V	5 V	
5 V	5 V	

3. Calculate the percentage of error.

MULTISIM based lab work:

1. Construct the Circuit 1 in MULTISIM.
2. Apply a **sine wave** with a peak-to-peak voltage = 2 V and frequency = 1 kHz to the input V_1.
3. Apply a **sine wave** with a peak-to-peak voltage = 3 V and frequency = 1 kHz to the input V_2.
4. Observe the input and output wave shapes through an oscilloscope.
5. Take readings.
6. $V_{o(p\text{-}p)}$ = _____

Questions:

1. Write the application of an adder circuit in audio mixing.
2. What are the advantages and disadvantages of an inverting adder circuit?
3. Design a 5-input inverting summing amplifier with a gain of $A = -3$. Show the equation of V_o.
4. Derive the output voltage expression of a 3-channel non-inverting weighted adder with superposition principle. What are the advantages and disadvantages of a non-inverting adder circuit?
5. Design a difference amplifier circuit with a differential gain of $A_d = 3$. Show the equation of V_o.

EEE 311, ETE 311, CSE 351 lab: Electronics II

Experiment No: 05

Name of the Experiment:

Integrator and Differentiator Circuits using Op-amp

Objective:

a. To study the op-amp integrator circuit.
b. To study the op-amp differentiator circuit.

Theory:

Op-amps can be used to perform mathematical operations like integration and differentiation.

- Integrator:

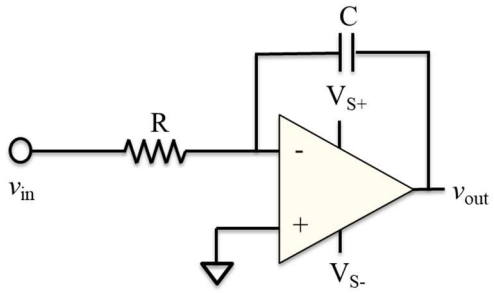


Figure 1: A basic integrator circuit.

Op-amp integrator can perform the mathematical operation of integration. The op-amp integrator produces an output voltage which is proportional to the integral of the input voltage. Hence, it can cause the output to respond to changes in the input voltage over time. The output v_{out} and input v_{in} can be related by the equation below:

$$v_{out} = -\frac{1}{RC}\int_0^t v_{in}\, dt$$

- Differentiator:

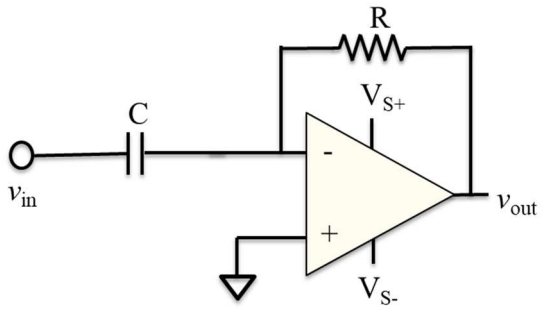


Figure 2: A basic differentiator circuit.

The working principle of an op-amp differentiator is the opposite to that of the integrator circuit. An op-amp differentiator performs the mathematical operation of differentiation. This circuit produces a voltage output which is directly proportional to the input voltage's rate-of-change with respect to time. The output v_{out} and input v_{in} can be related by the equation below:

$$v_{out} = -RC\frac{dv_{in}}{dt}$$

If a square wave is applied at the input of an integrator circuit, the output will be a triangular wave of the same frequency. Likewise, if a triangular wave is applied at the input of a differentiator circuit, the output will be a square wave of same frequency.

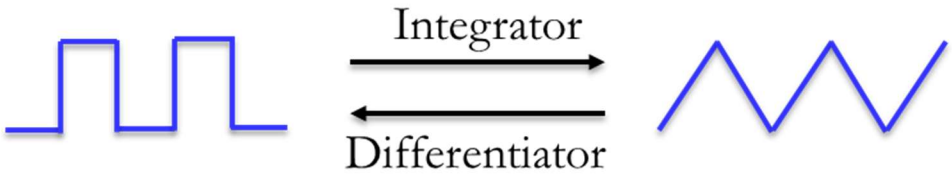
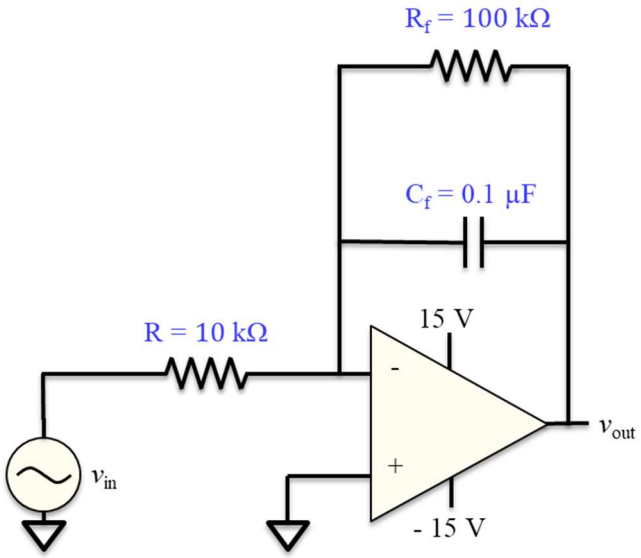


Figure 3: (a) Operation of an integrator circuit (b) Operation of a differentiator circuit.

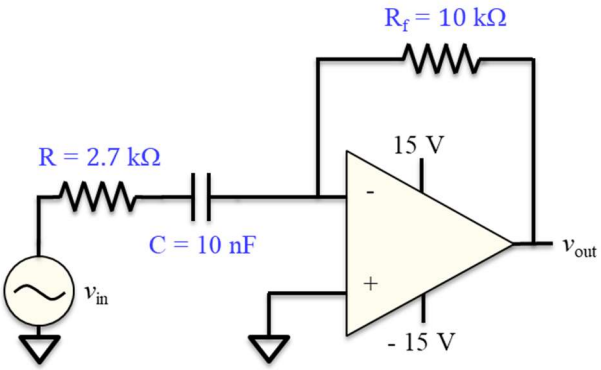
Equipment List:

	Component Details	Specification	Quantity
1.	Op-amp	LM741	1 piece
2.	Resistor	100 kΩ, 10 kΩ, 2.7 kΩ	1 piece of each
3.	Capacitors	0.1 µF, 10 nF	1 piece, 1 piece
4.	Oscilloscope	2 channels	1 unit
5.	Function Generator		1 unit
6.	Digital Multimeter		1 unit
7.	Breadboard		1 unit
8.	DC Power Supply		1 unit

Circuit Diagram:



Circuit 1 – Integrator Circuit

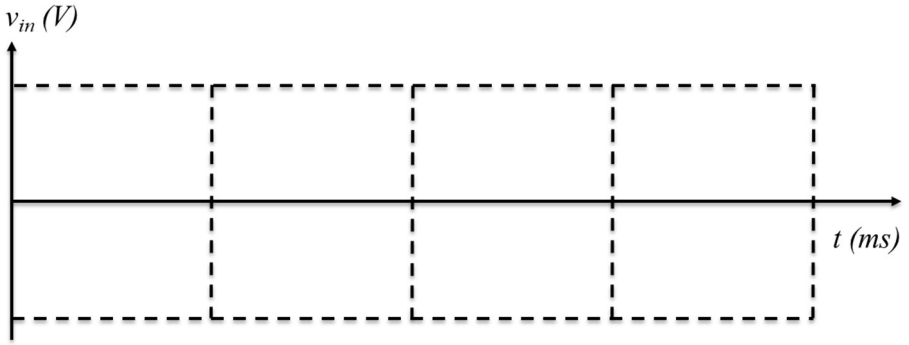


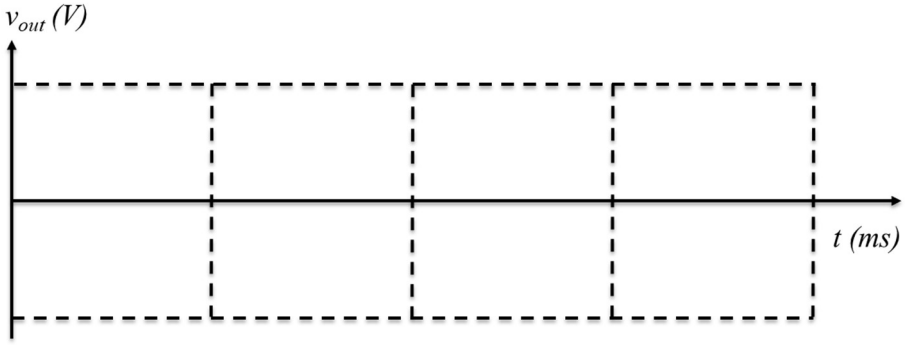
Circuit 2 – Differentiator Circuit

Procedure:

Integrator:

1. Construct circuit 1. Set V_{S+} = 15 V and V_{S-} = –15 V.
2. Apply a **sine wave** with a peak-to-peak voltage = 5 V and frequency = 1 kHz as the input of the integrator. Observe the output waveshape through an oscilloscope.
3. Take reading of its amplitude. $v_{in(p\text{-}p)}$ = _____, $v_{out(p\text{-}p)}$ = _____
4. Draw the input and output wave shapes with proper labeling.

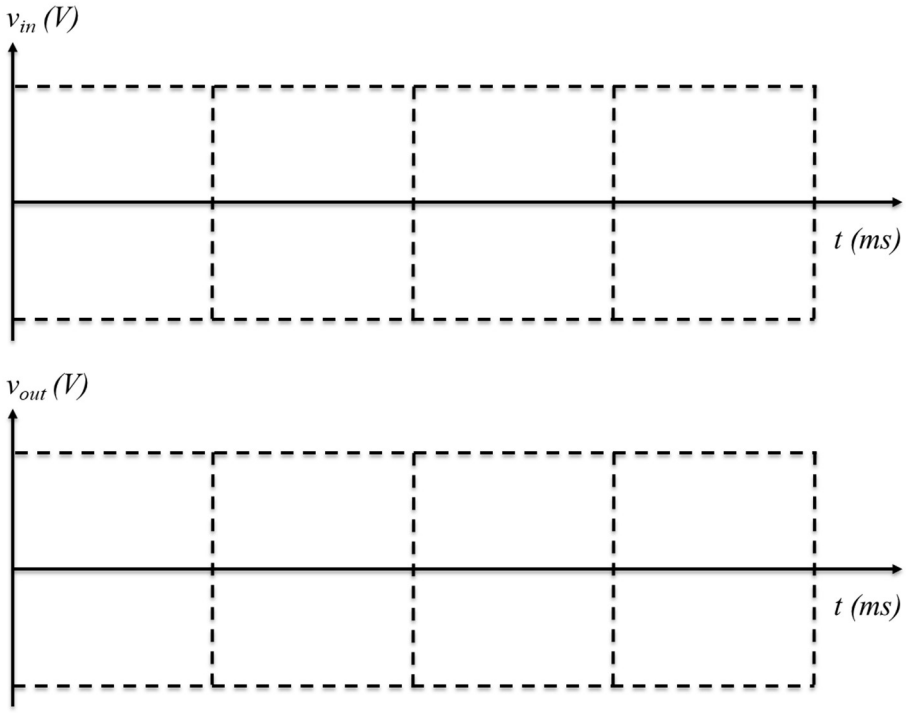




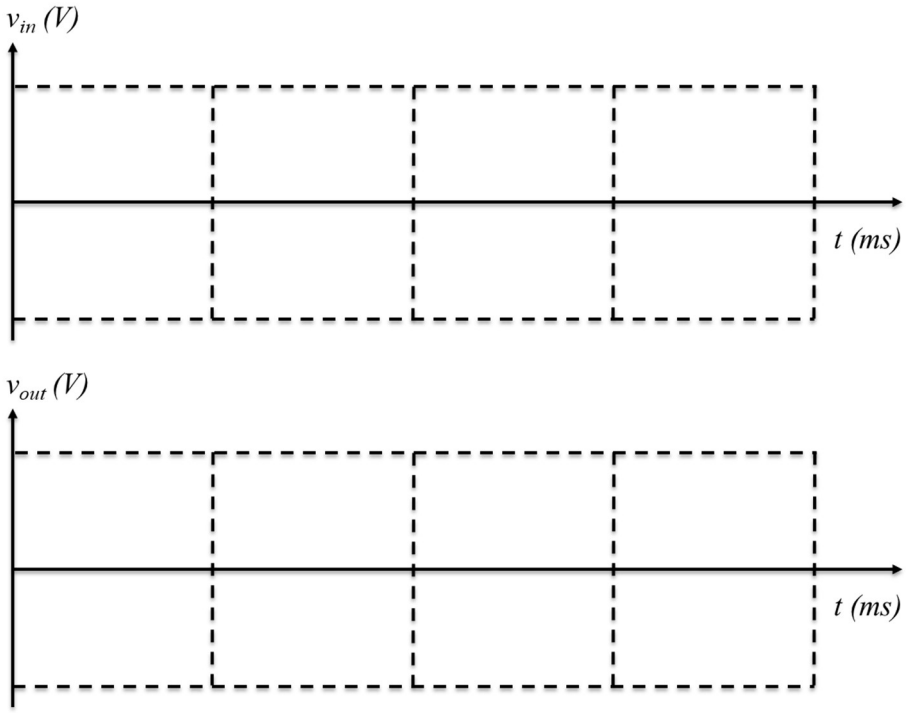
5. Now, apply a **square wave** with a peak-to-peak voltage = 5 V and frequency = 1 kHz

as the input of the integrator. Observe the output wave shape through an oscilloscope. This is the integrated output and it should be a triangular wave.

6. Take reading of its amplitude. $v_{in(p\text{-}p)}$ = _____, $v_{out(p\text{-}p)}$ = _____
7. Draw the input and output wave shapes with proper labeling.



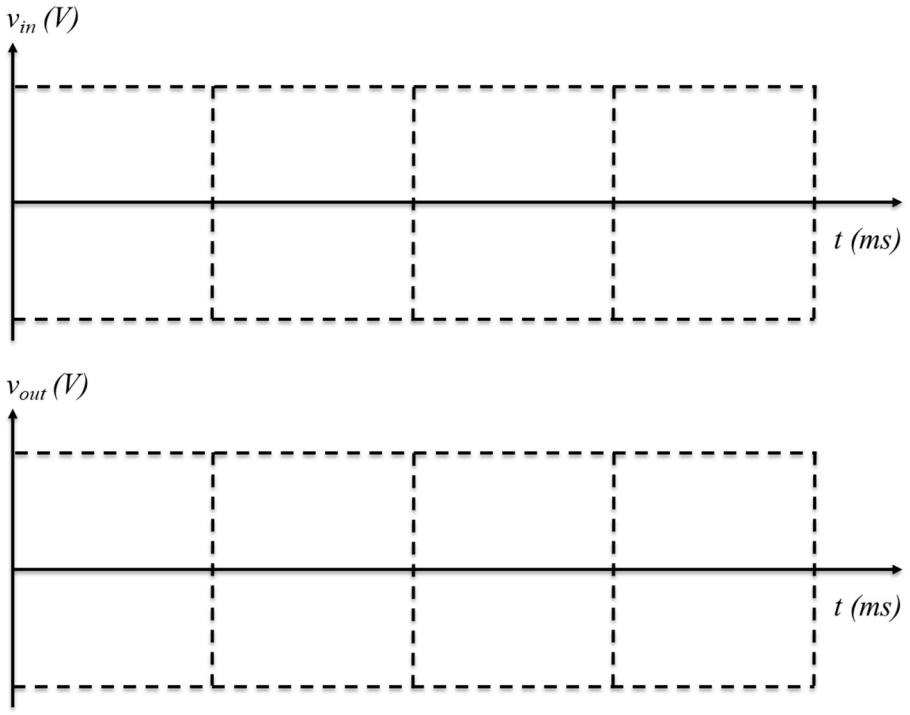
8. Now, apply a 4 V DC signal as the input of the integrator. Observe the output.
9. Take reading of its amplitude. v_{in} = _____, v_{out} = _____
10. Draw the input and output wave shapes with proper labeling.



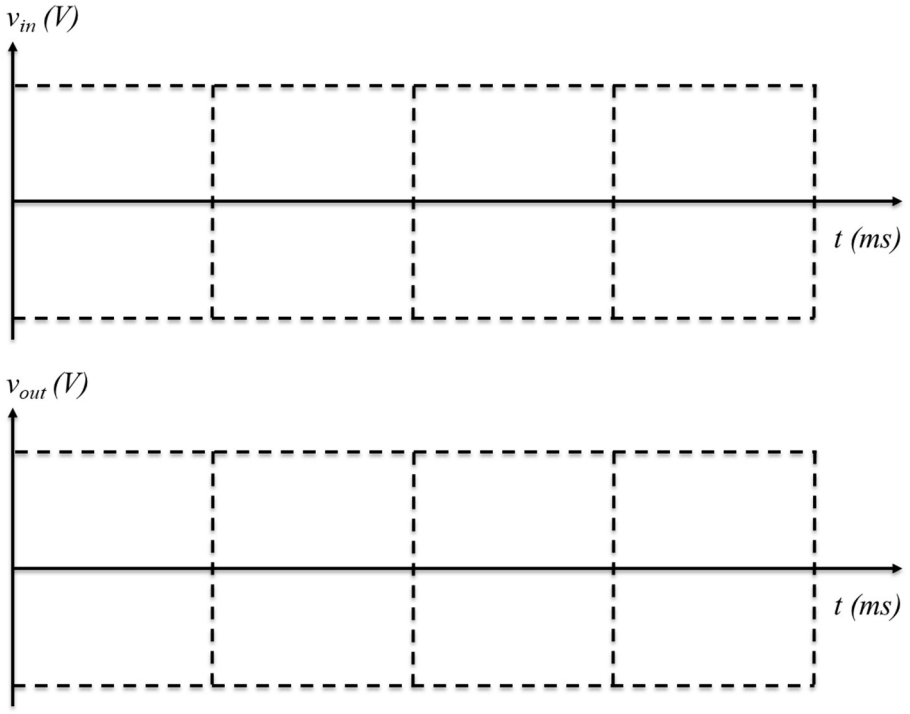
Differentiator:

1. Construct circuit 2. Set V_{S+} = 15 V and V_{S-} = –15 V.
2. Apply a **sine wave** with a peak-to-peak voltage = 5 V and frequency = 1 kHz as the input of the differentiator. Observe the output wave shape through an oscilloscope.

3. Take reading of its amplitude. $v_{in(p\text{-}p)}$ = _____, $v_{out(p\text{-}p)}$ = _____
4. Draw the input and output wave shapes with proper labeling.

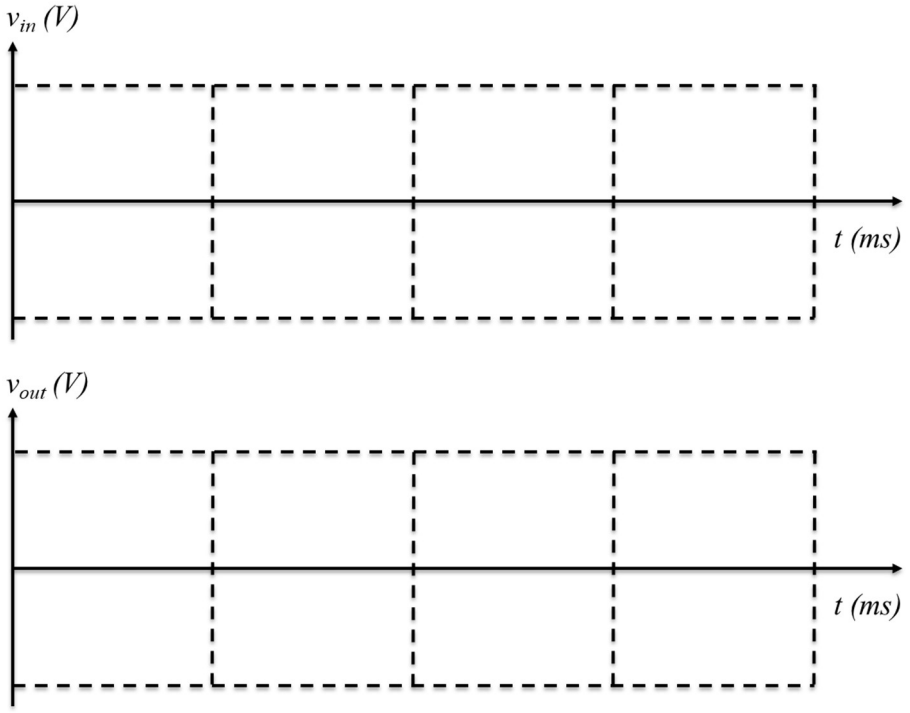


5. Now, apply a **triangular wave** with a peak-to-peak voltage = 5 V and frequency = 1 kHz as the input of the differentiator. Observe the output wave shape through an oscilloscope. This is the differentiated output and it should be a square wave.
6. Take reading of its amplitude. $v_{in(p\text{-}p)}$ = _____, $v_{out(p\text{-}p)}$ = _____
7. Draw the input and output wave shapes with proper labeling.



8. Now, apply a 4 V DC signal as the input of the differentiator. Observe the output.
9. Take reading of its amplitude. v_{in} = _____, v_{out} = _____

10. Draw the input and output wave shapes with proper labeling.



Questions:

1. What are the problems of basic integrator circuit? Draw an improved integrator circuit and show how it can overcome the drawbacks of the basic circuit.
2. What are the problems of basic differentiator circuit? Draw an improved differentiator circuit and show how it can overcome the drawbacks of the basic circuit.
3. Since the integrator and differentiator circuits contain capacitor, these circuits exhibit variations in gain with changing frequency of the input signal. Hence, these circuits work as filters. What type of filters are constructed from the following circuits?
    a. basic integrator,
    b. basic differentiator,
    c. practical integrator,
    d. practical differentiator.
4. What will be the output of an integrator circuit if the input is
    a. a constant DC signal,
    b. a sinusoidal wave?
5. What will be the output of a differentiator circuit if the input is
    a. a constant DC signal,
    b. a sinusoidal wave?

EEE 311, ETE 311, CSE 351 lab: Electronics II

Experiment No: 06

Name of the Experiment:

Study of First order Active High Pass Filter

Objective:

a. To study the basic construction of active high pass filter.
b. To study the frequency response curve of active high pass filter.

Theory:

Electronic filters

Electronic filters are electronic circuits which perform signal processing functions, specifically to remove unwanted frequency components from the signal, to enhance wanted ones, or both. Electronic filters can be:

- passive or active
- analog or digital

Passive filters

Passive implementations of linear filters are based on combinations of resistors (R), inductors (L) and capacitors (C). These types are collectively known as passive filters, because they do not depend upon an external power supply and/or they do not contain active components such as transistors.

Resistors on their own have no frequency-selective properties, but are added to inductors and capacitors to determine the time-constants of the circuit, and therefore the frequencies to which it responds. If an inductor is added to the design of a passive filter, the response curve will be closer to the ideal one. But there are some problems with inductors: they are bulky, costly and may have large internal resistive component.

The main disadvantage of passive filters is that the amplitude of the output signal is less than that of the input signal, the gain is never greater than unity and that the load impedance affects the filters characteristics. With passive filter circuits containing multiple stages, this loss in signal amplitude called "Attenuation" can become quiet severe. One way of restoring or controlling this loss of signal is by using amplification through the use of Active Filters.

Active filters

As their name implies, Active Filters contain active components such as operational amplifiers, transistors or FET's within their circuit design. They draw their power from an external power

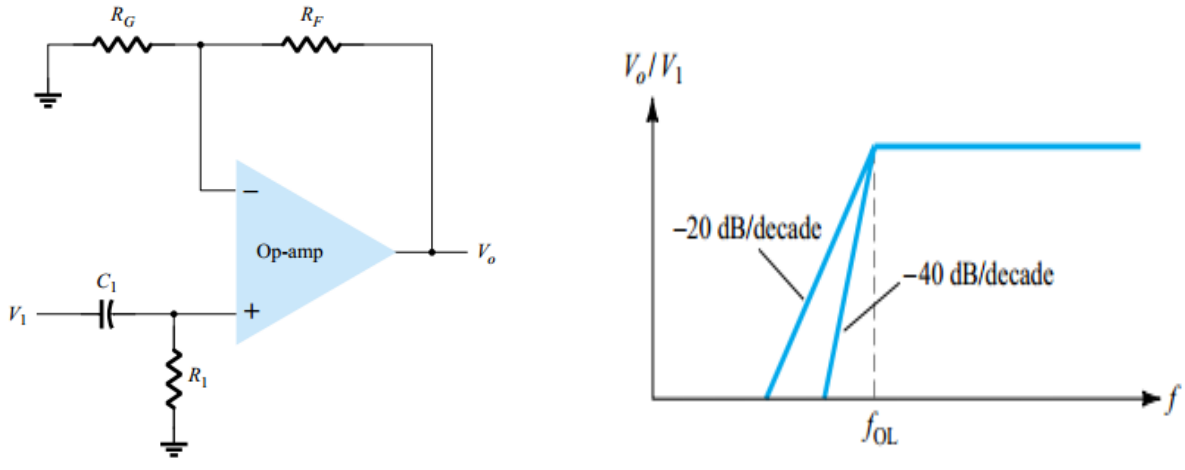
source and use it to boost or amplify the output signal. Filter amplification can also be used to either shape or alter the frequency response of the filter circuit by producing a more selective output response, making the output bandwidth of the filter narrower or even wider. Then the main difference between a passive filter and an active filter is amplification.

An active filter generally uses an operational amplifier (op-amp) within its design. Unlike a passive high pass filter which has in theory an infinite high frequency response, the maximum frequency response of an active filter is limited to the Gain/Bandwidth product (or open loop gain) of the operational amplifier being used. Still, active filters are generally easier to design than passive filters; they produce good performance characteristics, very good accuracy with a steep roll-off and low noise when used with a good circuit design. They are often smaller and less expensive because they do not require inductors. Inductors are bulky, they can't be used in IC. As active filers don't require inductors, IC realizations of them are feasible.

Active filters are implemented using a combination of passive and active components, and require an outside power source. High-pass filter, low-pass filter, band-pass filter, band reject filter and notch filter are some of the variants.

A high-pass filter is an electronic filter that passes signals with a frequency higher than a certain cutoff frequency and attenuates signals with frequencies lower than the cutoff frequency. The amount of attenuation for each frequency depends on the filter design. A high-pass filter is usually modeled as a linear time-invariant system. It is sometimes called a low-cut filter or bass-cut filter.

The basic operation of an Active High Pass Filter (HPF) is the same as for its equivalent RC passive high pass filter circuit, except this time the circuit has an op amp or transistor included within its design providing amplification and gain control. High-pass filters have many uses such as: blocking DC from circuitry sensitive to non-zero average voltages or radio frequency devices. They are used as part of an audio crossover to direct high frequencies to a tweeter while attenuating bass signals which could interfere with, or damage, the speaker. High-pass and low-pass filters are also used in digital image processing to perform image modifications, enhancements, noise reduction, etc., using designs done in either the spatial domain or the frequency domain



(a) (b)

Figure 1: (a) Non-inverting configuration, (c, d) Inverting configuration

Filter Order

The inductors and capacitors are the reactive elements of the filter. The number of reactive elements determines the order of the filter. In applications that use filters to shape the frequency spectrum of a signal such as in communications or control systems, the shape or width of the roll-off also called the "transition band", for a simple first-order filter may be too long or wide and so active filters designed with more than one "order" are required. These types of filters are commonly known as "High-order" or "n^{th}-order" filters. The complexity or filter type is defined by the filters "order", and which is dependent upon the number of reactive components such as capacitors or inductors within its design.

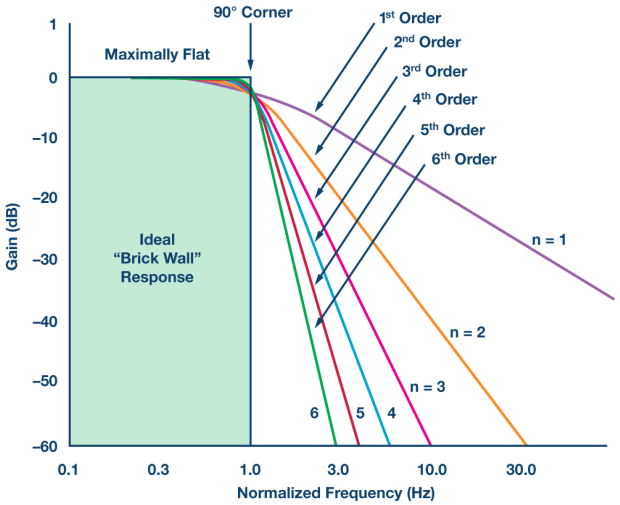
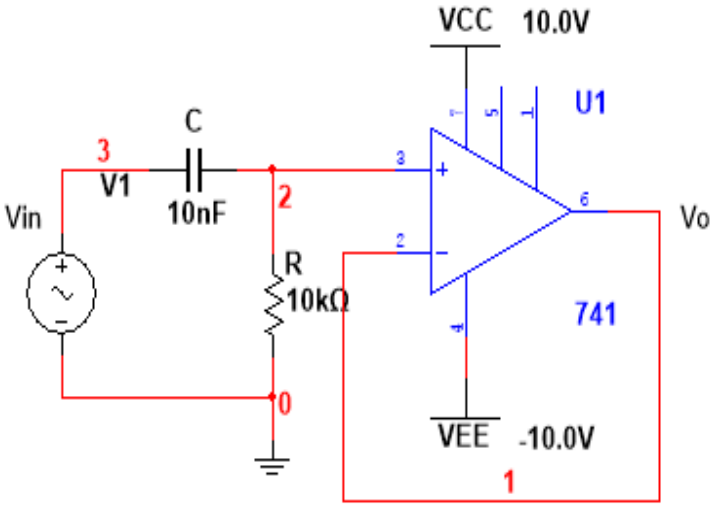


Figure 2: Frequency response of low pass filter of different orders

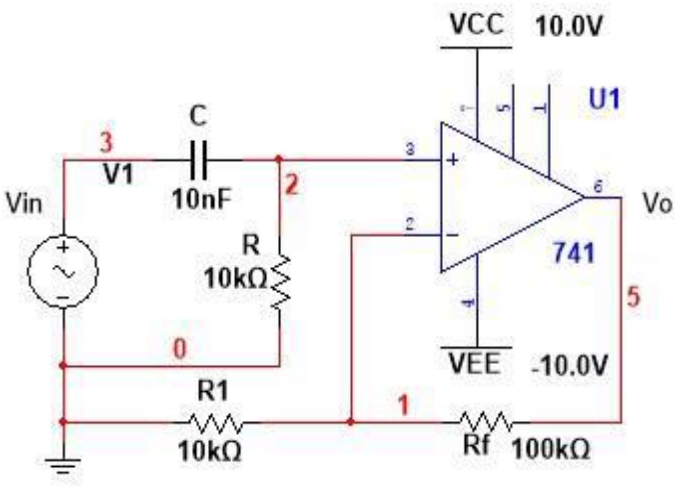
Equipment List:

	Component Details	Specification	Quantity
1.	Op-Amp	LM741	2 pieces
2.	Resistor	100kΩ, 10kΩ	1 piece, 2 pieces
3.	Capacitors	10nF	1 piece
4.	Oscilloscope		1 unit
5.	Function Generator		1 unit
6.	Digital Multimeter		1 unit
7.	Breadboard		1 unit
8.	DC Power Supply		1 unit

Circuit Diagram:



Circuit 1 – Active high pass filter circuit with gain 1



Circuit 2 – Active high pass filter circuit with gain 11

Procedure:

1. Calculate the cutoff frequency f_c using the equation below

$$f_c = \frac{1}{2\pi RC} =$$

2. Construct the circuit of circuit 1. Set $+V_{CC}$ = 10 V and $-V_{EE}$ = –10 V.
3. Apply a sine wave with a peak-to-peak voltage = 5 V as the input.
4. Vary the input frequency while keeping the same amplitude and observe the output wave shape through an oscilloscope. Record the output voltage in the data table.
5. Take at least 15 data until the cutoff frequency and then there is a flat region (take 10 data). Finally after the flat region, output voltage reduces significantly (take 10 data till 5 MHz).
6. Draw the response curve (gain vs. frequency) in a semi log graph paper and find the 3-db frequency or cut-off frequency from the graph.
7. Now, construct the circuit of circuit 2. Set $+V_{CC}$ = 10 V and $-V_{EE}$ = –10 V.
8. Apply a sine wave with a peak-to-peak voltage = 1 V as the input.
9. Repeat procedure no. 4, 5 & 6
10. Plot both frequency response curves on same log graph.
11. What is the practical cutoff frequency?

$$f_{c(practical)} =$$

Questions:

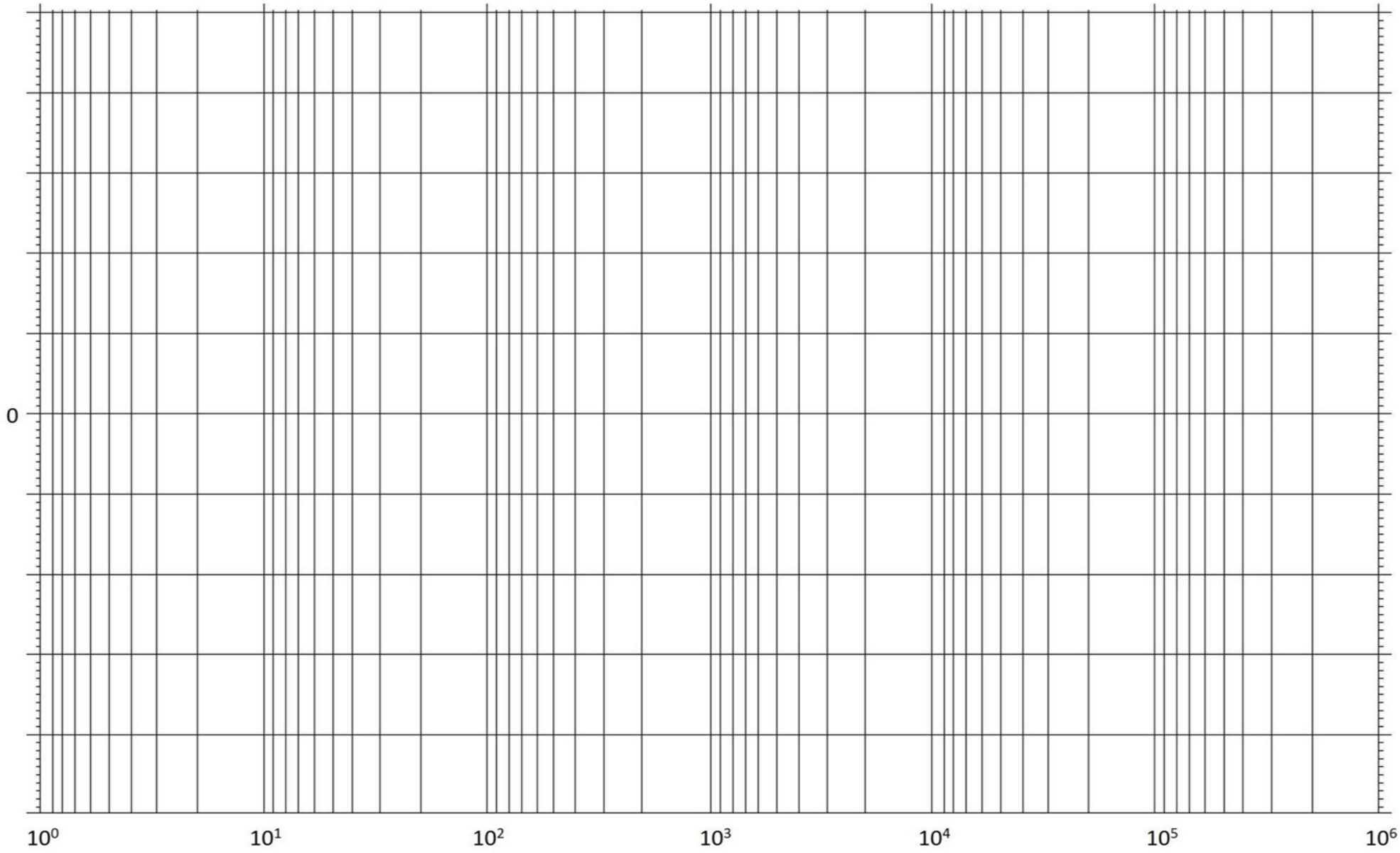
1. What is the definition of active and passive elements in electronics? Give examples.
2. Design a band pass and band stop filter using low pass and high pass filter. Show your design in both block diagram and circuit diagram.
3. What defines the order of a filter?
4. Why we use capacitor rather than inductor in active filters?
5. What is the advantages of using Op-Amp in high pass filter?
6. You have a series circuit consisting of an AC power supply, an 1150.0 ohm resistor and a 144.0nf capacitor. If the circuit is configured as an Active high pass filter, what frequency will cause the gain to be 0.25?

Data Sheet: (Gain 1)

Input Frequency (Hz)	$V_{in\ (p-p)}$	$V_{out\ (p-p)}$	Gain (Vo / Vi)	Gain (dB) = 20 Log (Vo/Vi)

Data Sheet: (Gain 11)

Input Frequency (Hz)	$V_{in\ (p-p)}$	$V_{out\ (p-p)}$	Gain (Vo / Vi)	Gain (dB) = 20 Log (Vo/Vi)


0
10^0
10^1
10^2
10^3
10^4
10^5
10^6

EEE 311, ETE 311, CSE 351 lab: Electronics II

Experiment No: 07

Name of the Experiment:

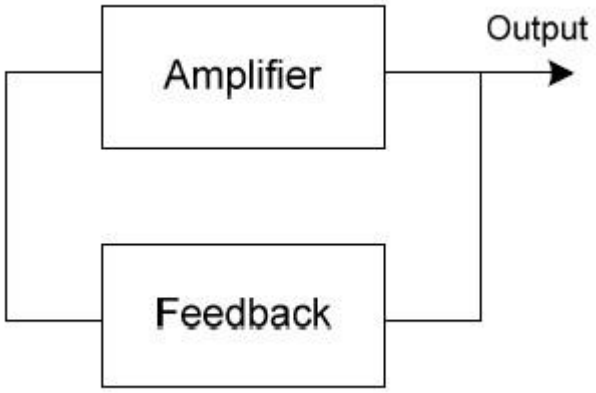
Designing a Wien Bridge Oscillator using Op-Amp

Objectives:

a. To study the basic operations of Wien Bridge oscillator
b. To find out the resonance frequency and feedback gain of a Wien Bridge oscillator

Theory:

An oscillator is a circuit that produces periodic electric signals such as sine wave or square wave. The application of oscillator includes sine wave generator, local oscillator for synchronous receivers etc. An oscillator consists of an active device and a feedback network. Active device i.e. op-amp is used as an amplifier. Passive components such as R-C or L-C combinations are used as feedback network.

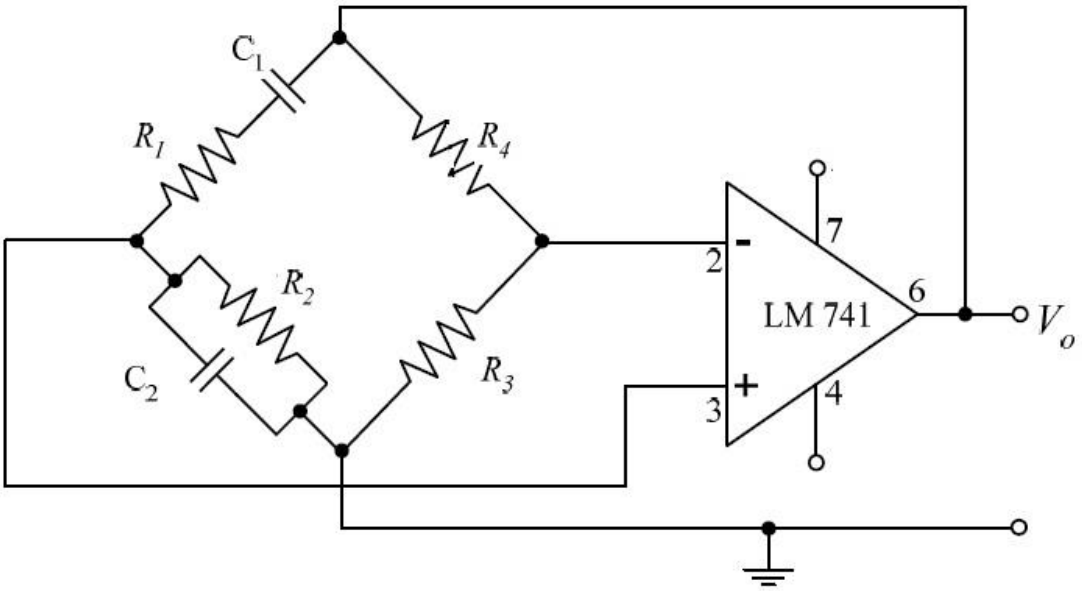


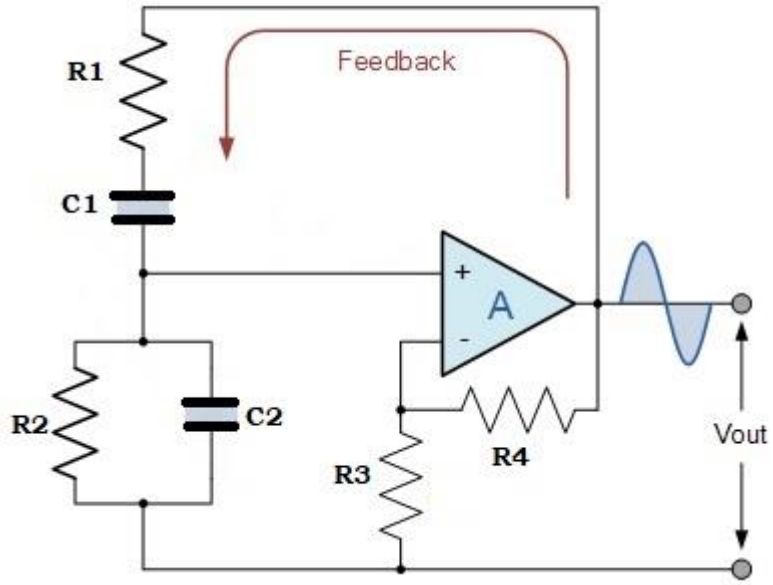
To start the oscillation with the constant amplitude, positive feedback is not the only sufficient condition. Oscillator circuit must satisfy the following two conditions known as **Barkhausen** conditions:

1. Magnitude of the loop gain ($A_v \beta$) = 1
Here, A_v = Amplifier gain and β = Feedback gain.
2. Phase shift around the loop must be 360° or 0°.

One of the simplest sine wave oscillators which uses a RC network to produce a sinusoidal output waveform, is called a Wien Bridge Oscillator. Wien bridge oscillator is an audio frequency sine wave oscillator of high stability and simplicity.

The feedback signal in this circuit is connected to the non-inverting input terminal so that the op-amp is working as a non-inverting amplifier. Therefore, the feedback network does not provide any phase shift. The circuit can be viewed as a Wien bridge with a series combination of R1 and C1 in one arm and parallel combination of R2 and C2 in the adjoining arm. Resistors R3 and R4 are connected in the remaining two arms. The condition of zero phase shift around the circuit is achieved by balancing the bridge.





At high frequencies, the reactance of capacitor C_1 and C_2 approaches zero. This causes C_1 and C_2 to appear as a short. Here, capacitor C_2 shorts the resistor R_2. Hence, the output voltage V_o will be zero since the voltage at the non-inverting input is equal to the voltage drop across R_2 and C_2 combination.

At low frequencies, both capacitors act as open because capacitor offers very high reactance. Again, output voltage will be zero because the input signal is dropped across the R_1 and C_1 combination.

But at one particular frequency between the two extremes, the output voltage reaches to the maximum value. At this frequency only, resistance value becomes equal to capacitive reactance and

gives maximum output. Hence, this frequency is known as oscillating frequency (f). The ideal frequency of oscillation is:

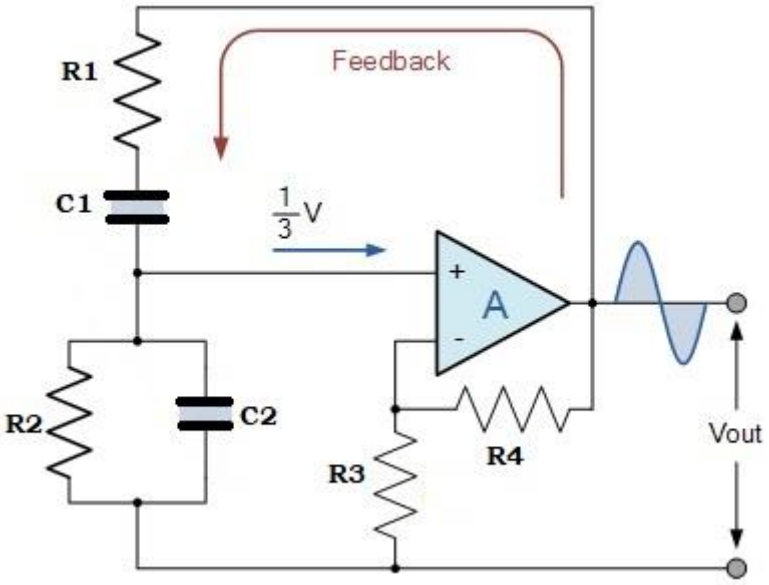
$$f_o = \frac{1}{2\pi\sqrt{R_1 C_1 R_2 C_2}}$$

At the resonance frequency, feedback gain,

$$\beta = \frac{R_2 C_1}{R_1 C_1 + R_2 C_1 + R_2 C_2}$$

We have to take A_v such that $A_v \beta$ is equal to 1 or greater than 1.

If we set $R_1 = R_2$ and $C_1 = C_2$, we get, $\beta = {}^1/_3$

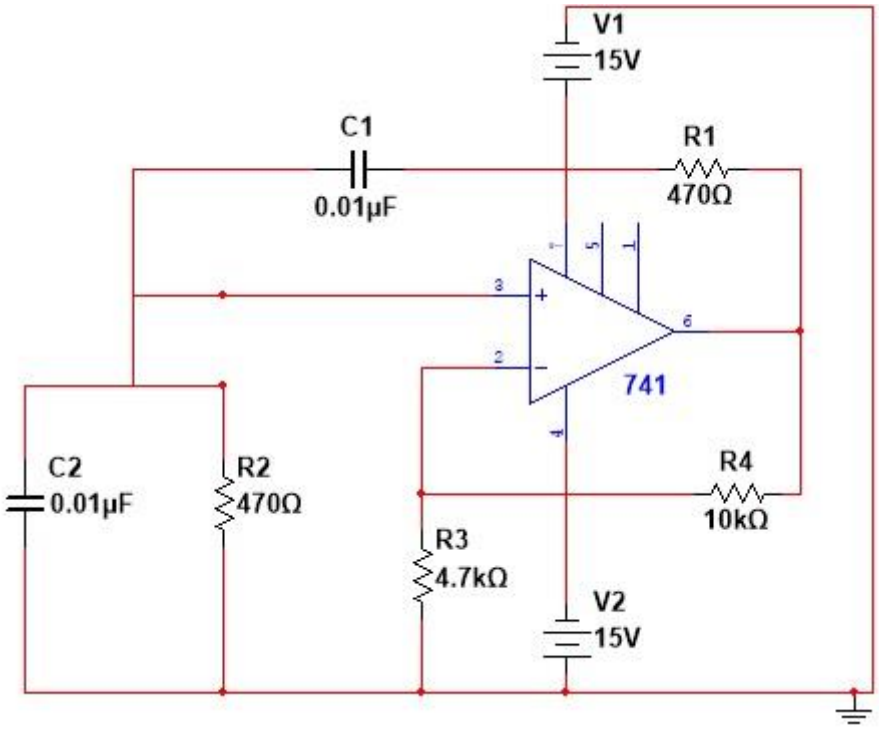


So, $A_v = 1 + \frac{R_4}{R_3}$ has to be equal or greater than 3.

Equipment List:

Serial No.	Component Details	Specification	Quantity
1.	Op-Amp	LM 741	1 piece
2.	Resistor	10kΩ, 4.7kΩ, 470 Ω	1 piece, 1 pieces, 2 pieces
3.	Capacitor	0.01 µF	2 pieces
4.	Oscilloscope		1 unit
5.	DC power supply		2 units
6.	Digital Multimeter		1 unit
7.	Bread Board		1 unit

Circuit Diagram:



Procedure:

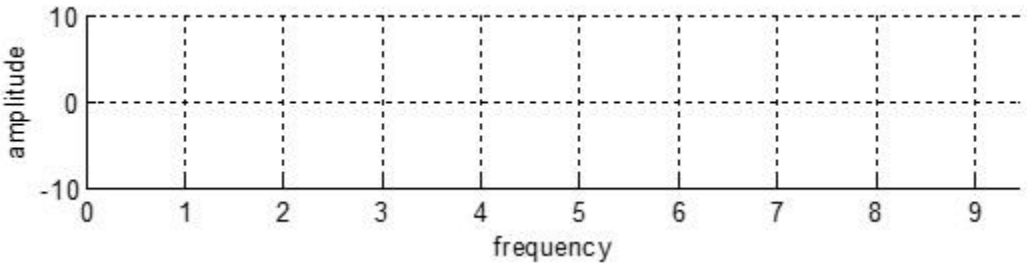
1. Connect the circuit as shown in the figure.
2. Observe the output on oscilloscope. Draw the waveform.
3. Take the readings of amplitude and frequency of the output waveform.
   Amplitude (peak to peak) = _____, Frequency = _____
4. Calculate the theoretical resonance frequency and compare it with the practical one.

Theoretical resonance frequency	Practical resonance frequency
$f_o = \frac{1}{2\pi\sqrt{R_1 C_1 R_2 C_2}} =$	$f_o =$

5. Calculate the percentage of error.

$$error = \frac{|\ theoretical\ value - practical\ value\ |}{theoretical\ value} \times 100 = _____\%$$

Graph:



Questions:

1. Write down the names of five electrical oscillators. What are the main advantage(s) of Wien Bridge oscillator over other oscillators?
2. What will happen if we change the value of components inside the RC feedback circuit?
3. What is the feedback gain of the circuit we designed in our lab?
4. Design a Wien Bridge oscillator that will provide a sine wave of 3 kHz.
5. In the circuit that you have designed in question no.4, how can you change the output signals amplitude without changing the frequency?

Experiment No: 6

Study of Bi-Stable Operation & Voltage Controlled Oscillator (VCO)

Objectives:

a. To observe the basic operation and the output of Bi-stable mode.
b. To observe the operation and output of Voltage Controlled Oscillator (VCO).

Theory:

The name of the timer comes from the three 5 kΩ resistors which are embedded in it. This IC gives precise time at the output which is must in the time related circuits. Its basic operations are to produce clock pulses with predefined frequency as an astable, mono-stable & bi-stable mutivibrator. We will see these three operations in this experiment. The following figures are the pin configuration & layout of the 555 Timer IC as which allows us to focus on the functions of the circuit.

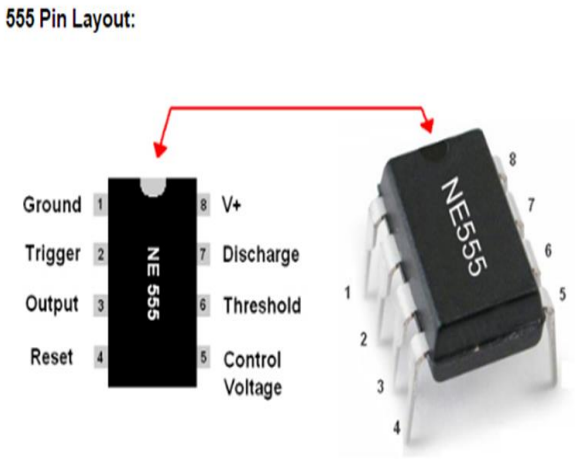


Figure 1: Pin configuration of 555 Timer.

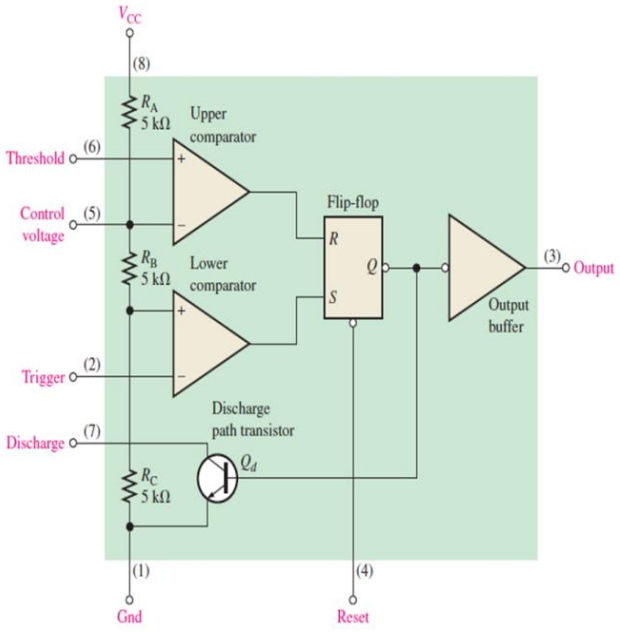


Figure 2: Internal Pin Configuration of 555 Timer

Bi-stable Mode:

Bi-stable Multivibrator mode of 555 timer IC is the easiest mode of 555 timer IC, where Monostable multibrator mode has one stable & one unstable state, Astable multvibrator mode has both the unstable states, here in Bi-stable mode, both the states are stable. Means it remains in the same state (either HIGH or LOW) until an external trigger is applied; otherwise it stays in

one of the two states (HIGH or LOW) indefinitely. In bi-stable mode there is no RC network like the other two modes of 555, hence there are no equations and wave form. Bi-stable mode simply works as a Flip-flop.

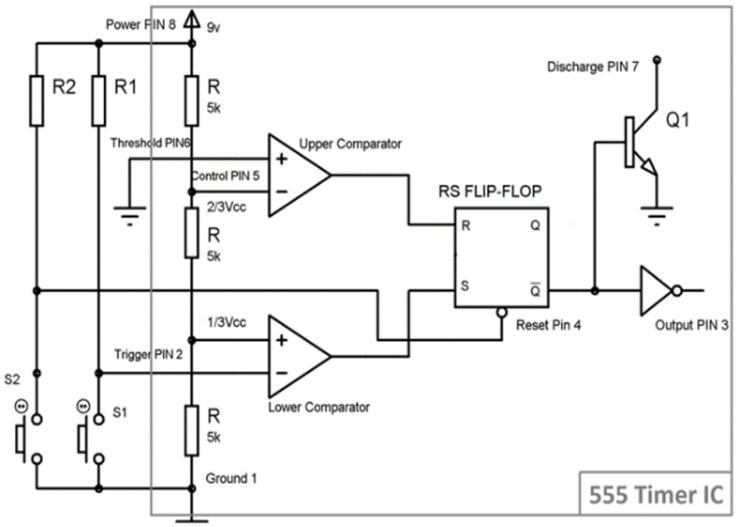


Fig 3: 555 Timer as Bi-stable Mode.

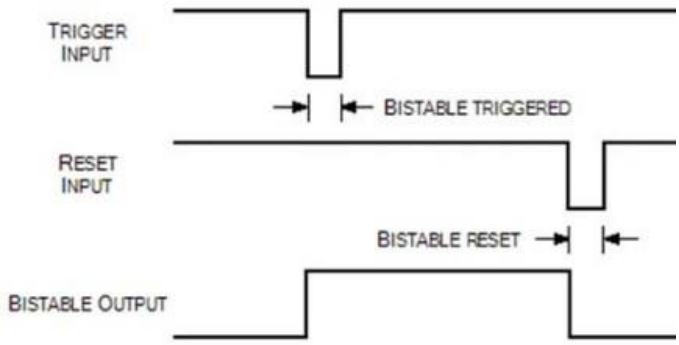


Fig 4: Timing diagram for Bi-stable Mode

Voltage Controlled Oscillator:

The 555 timer consists basically of two comparators, a flip-flop, a discharge transistor, and a resistive voltage divider, as shown in Figure. The flip-flop (bi-stable multivibrator) is a digital device that may be unfamiliar to you at this point unless you already have taken a digital fundamentals course. Briefly, it is a two-state device whose output can be at either a high voltage level (set, S) or a low voltage level (reset, R). The state of the output can be changed with proper input signals. The resistive voltage divider is used to set the voltage comparator levels. All three resistors are of equal value; therefore, the upper comparator has a reference of $2/3V_{CC}$, and the lower comparator has a reference of $1/3V_{CC}$. The comparators' outputs control the state of the flip-flop. When the trigger voltage goes below $1/3V_{CC}$, the flip-flop sets and the output jumps to its high level. The threshold input is normally connected to an external RC timing circuit. When the external capacitor voltage exceeds $2/3V_{CC}$, the upper comparator resets the flip-flop, which in turn switches the output back to its low level. When the device output is low, the discharge transistor is turned on and provides a path for rapid discharge of the external timing capacitor. This basic operation allows the timer to be configured with external components as an oscillator, a one-shot, or a time delay element. A 555 timer can be set up to operate as a VCO by using the same external connections as for astable operation, with the exception that a variable control voltage is applied to the CONT input (pin 5), as indicated in Figure. As shown in Figure 16–43, the control voltage changes the threshold values of $1/3V_{CC}$ and $2/3V_{CC}$ for the internal comparators. With the control voltage, the upper value is and the lower value is $1/2V_{CONT}$, as you can see by examining the internal diagram of the 555 timer. When the control voltage is varied,

the output frequency also varies. An increase in increases the charging and discharging time of the external capacitor and causes the frequency to decrease. A decrease in decreases the charging and discharging time of the capacitor and causes the frequency to increase. An interesting application of the VCO is in phase-locked loops, which are used in various types of communication receivers to track variations in the frequency of incoming signals.

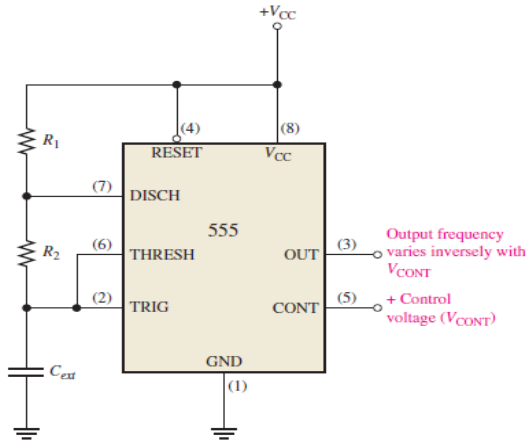


Figure 5: Voltage Control Oscillator (VCO)

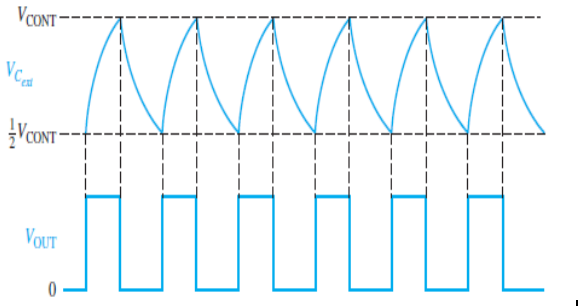


Figure 6: Timing diagram for VCO

Equipment List:

Serial Number	Component Detail	Specification	Quantity
1	Resistor	10KΩ (x 2), 330Ω	3 piece
2	Capacitor	10nF, 100nF	2 pieces
3	Push Switch		2 pieces
4	555 Timer	LM555	1 piece
5	Oscilloscope		1 unit
6	DMM		1 unit
7	DC Voltage Source		2 units
8	Bread Board		1 unit

Circuit Diagram:

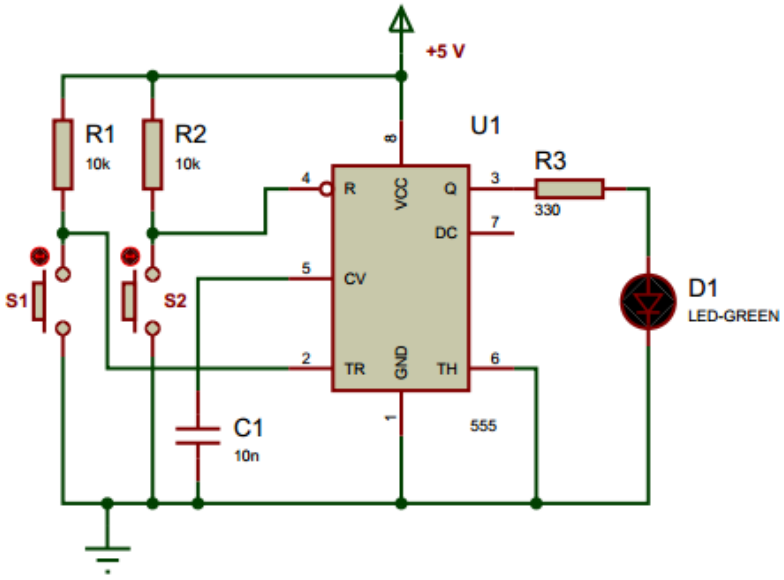


Figure 7: Bi-stable mode for 555 timer.

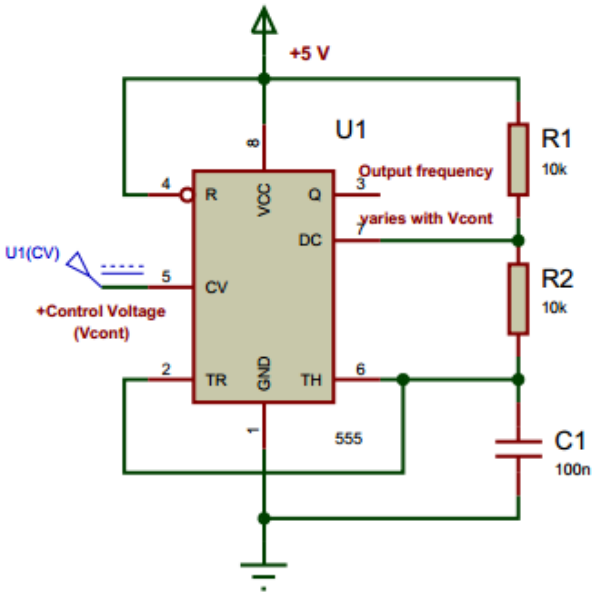


Figure 8: VCO circuit diagram of 555 timer.

Procedure:

1. Construct the Figure: 7 on the breadboard.
2. Press switch 1 and check the LED. Measure voltage at pin 3.
3. Press switch 2 and check the LED. Measure voltage at pin 3.
4. Construct the Figure: 8 on the breadboard.
5. Apply control voltage source at pin 5 and oscilloscope at pin 3.
6. Record the time duration and magnitude of output from oscilloscope.
7. Draw the wave shape of the output and compare with its reference one.

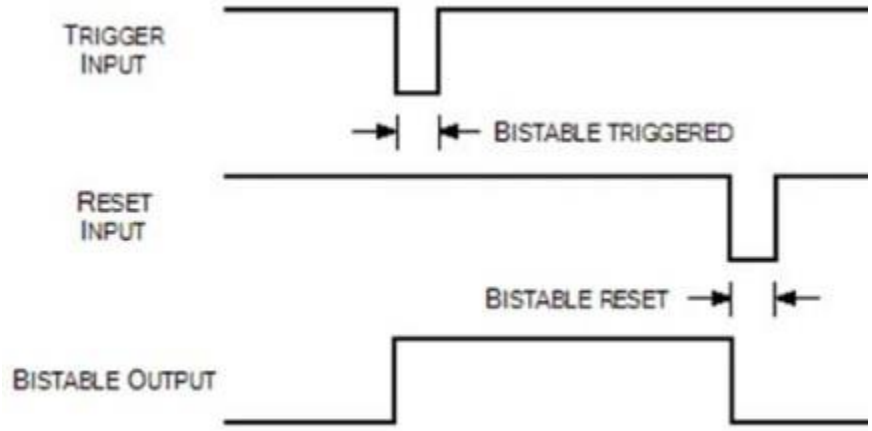
Result and Findings:

After completing the experiment, the student will take the wave-shapes found in the experiment. They will take all the numerical data like magnitude of the output, time/duration as high, time/duration as low. They will match the result with the formulae.

Data Sheet:

Bi-stable circuit

	Nominal Value		Measured Value
V_{CC}	= _____V	V_{CC}	= _____V
R_1	= _____ k Ω	R_1	= _____ k Ω
R_2	= _____ k Ω	R_2	= _____ k Ω
C_1	= _____ µF	C_1	= _____ µF



Record the output voltage at pin 3

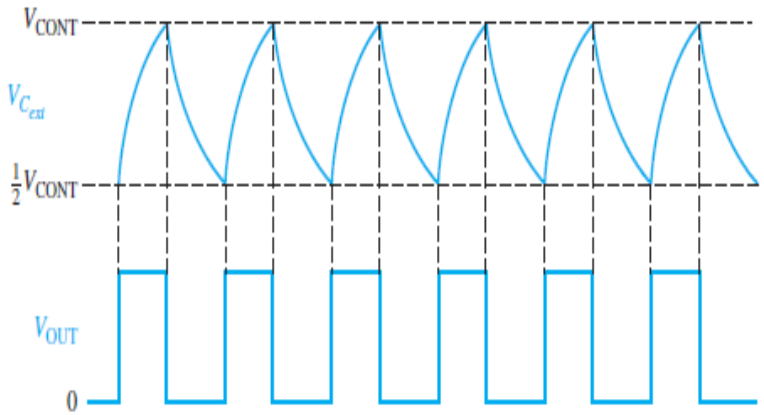
When LED is on: V_{out} = _____

When LED is off: V_{out} = _____

Data Sheet:

VCO circuit

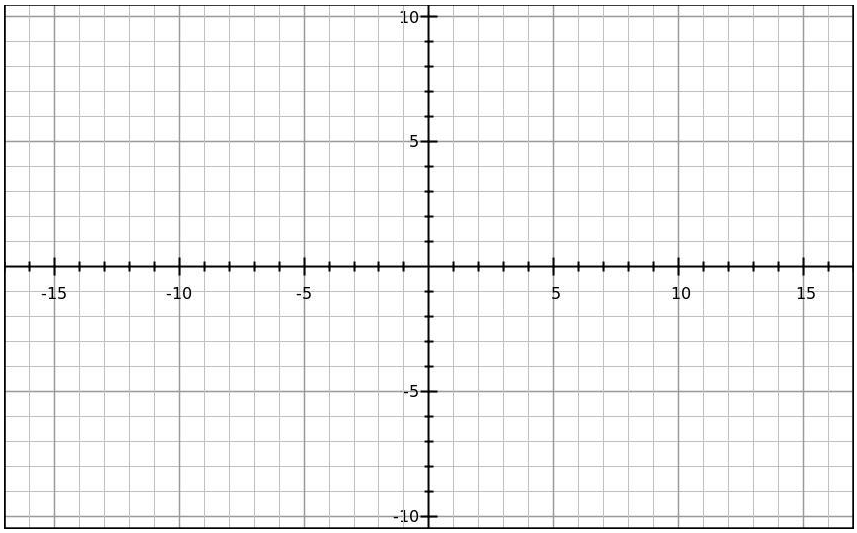
Nominal Value		Measured Value	
V_{CC}	= _____V	V_{CC}	= _____V
R_1	= _____ k Ω	R_1	= _____ k Ω
R_2	= _____ k Ω	R_2	= _____ k Ω
R_3	= _____ k Ω	R_3	= _____ k Ω
C_1	= _____ μF	C_1	= _____ μF



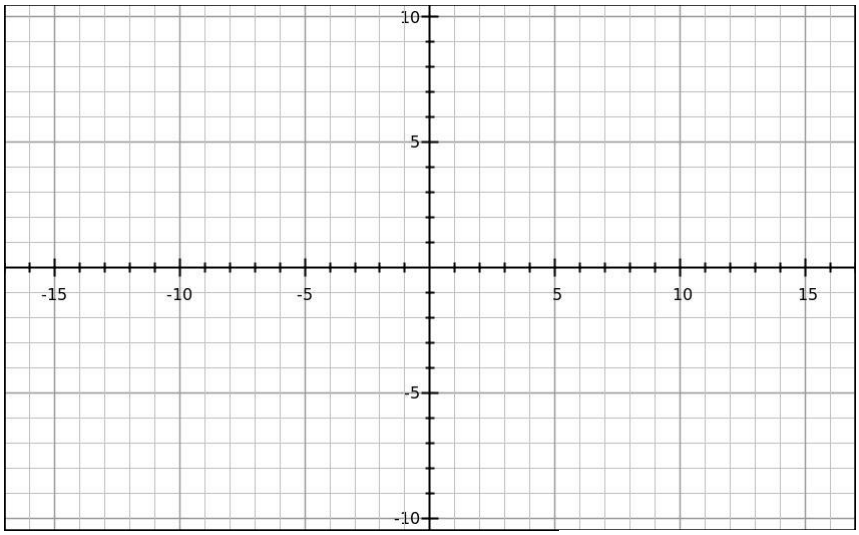
Record the values and complete the table using Experimental Data

V_{cont} (V)	$V_{out\text{-}pp}$	T_{out}	f_{out}	Duty cycle
1				
2				
2.5				
3.0				
3.5				
4.0				
4.5				

Bi-stable Circuit: Draw Bi-stable Output, Trigger input and Reset input



VCO Circuit: Draw V_c and V_{out}



Report Question:

1. What are the differences between astable, mono-stable, and bi-stable multivibrator?
2. What are the two reference voltage when V_{cc} = 10V.
3. Determine the frequency of oscillation for the 555 astable oscillator in the following figure.
4. What will the value of C_{ext} when we change the frequency to 25 KHz?

